



WHEN TRUST MATTERS

# Maritime cyber security –IACS UR and DNV Cyber secure

April 2024

Manufacturer seminar - Korea



# DNV Cyber secure class notation & the IACS UR

# New IACS Unified Requirements on Cyber Security supports Classification Societies' role in safeguarding life and vessels

**New vessels contracted after 1<sup>st</sup> of July 2024 will need to be approved according to Cyber security**

**The requirements come to yards, designers and owners (UR E26) as well as suppliers (UR E27)**

**Majority of merchant vessels above 500GT in international trade will be impacted**

**Safety related systems such as automation, navigation, fire and communication to be approved**



## **IACS adopts new requirements on cyber safety**

Recognising that cyber incidents on vessels can have a direct and detrimental impact on life, property, and the environment, IACS has steadily increased its focus on the reliability and functional effectiveness of onboard, safety-critical, computer-based systems.

IACS identified at an early stage that, for ships to be resilient against cyber incidents, all parts of the industry needed to be actively involved, and so convened a Joint Working Group (JWG) on Cyber Systems which helped identify best practices, appropriate existing standards in risk and cyber security, and a practical risk-based approach.

Building on this extensive collaboration, and utilising the experience gained from its existing Recommendations, as well as developments at IMO including, in particular, IMO Resolution MSC.428(98) applicable to in-service vessels since the 1<sup>st</sup> of Jan 2021, IACS has adopted two new IACS Unified Requirements (URs) on the cyber resilience of Ships:

**UR E26** aims to ensure the secure integration of both Operational Technology (OT) and Information Technology (IT) equipment into the vessel's network during the design, construction, commissioning, and operational life of the ship. This UR targets the ship as a collective entity for cyber resilience and covers five key aspects: equipment identification, protection, attack detection, response, and recovery.

**UR E27** aims to ensure system integrity is secured and hardened by third-party equipment suppliers. This UR provides requirements for cyber resilience of onboard systems and equipment and provides additional requirements relating to the interface between users and computer-based systems onboard, as well as product design and development requirements for new devices before their implementation onboard ships.

# Vessels and systems in scope for the IACS UR on cyber security

## 1.3.1 Vessels in scope

This UR is applicable to the following vessels:

- Passenger ships (including passenger high-speed craft) engaged in international voyages
- Cargo ships of 500 GT and upwards engaged in international voyages
- High speed craft of 500 GT and upwards engaged in international voyages
- Mobile offshore drilling units of 500 GT and upwards
- Self-propelled mobile offshore units engaged in construction (ie wind turbine installation maintenance and repair, crane units, drilling tenders, accommodation, etc)

This UR may be used as non-mandatory guidance to the following.

- Ships of war and troopships
- Cargo ships less than 500 gross tonnage
- Vessels not propelled by mechanical means
- Wooden ships of primitive build
- Passenger yachts (passengers not more than 12)
- Pleasure yachts not engaged in trade
- Fishing vessels
- Site specific offshore installations (ie FPSOs, FSUs, etc)

**Table 12 Default SuC**

| <i>Function</i>                                                                                                                                                                                                                  | <i>Examples of systems <sup>1)</sup></i>                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Propulsion                                                                                                                                                                                                                       | Systems needed for propulsion of the vessel. E.g. propeller pitch, engine speed, thruster, electrical motors and drives.                                                                                                                                                                                                               |
| Steering                                                                                                                                                                                                                         | Systems needed for steering of the vessel. E.g. rudder, azimuth thruster, electrical motors and drives.                                                                                                                                                                                                                                |
| Watertight integrity                                                                                                                                                                                                             | Systems needed to meet requirements for internal and external watertight integrity. E.g. internal watertight doors, shell doors, flooding detection.                                                                                                                                                                                   |
| Bilge, ballast                                                                                                                                                                                                                   | Systems needed to meet requirements for bilge drainage, and ballasting. E.g. valves, pumps, level measurement, loading computer                                                                                                                                                                                                        |
| Anchoring and mooring                                                                                                                                                                                                            | System(s) needed to meet requirements for anchoring and mooring in <a href="#">Pt.3 Ch.11 Sec.1</a> .                                                                                                                                                                                                                                  |
| Electrical power                                                                                                                                                                                                                 | Systems needed for generation and distribution of electrical power. E.g. main and auxiliary engine, generator, switchboard, energy storage system, power management.                                                                                                                                                                   |
| Auxiliary systems                                                                                                                                                                                                                | Systems needed to serve required auxiliary functions. E.g. cooling, fuel, lubrication, compressed air, ventilation.                                                                                                                                                                                                                    |
| Safety functions                                                                                                                                                                                                                 | Safety systems required by applicable rules for classification. E.g. emergency shutdown, machinery protection, boiler safety, ignition source control, cargo tank overfill protection, gas detection system.                                                                                                                           |
| Lighting                                                                                                                                                                                                                         | System needed to control main, emergency and navigation lights.                                                                                                                                                                                                                                                                        |
| Fire safety                                                                                                                                                                                                                      | Fire detection and alarm system and fixed fire extinguishing system required by applicable statutory regulations or rules for classification.                                                                                                                                                                                          |
| Navigation                                                                                                                                                                                                                       | Navigational systems and equipment required by applicable statutory regulations (e.g. SOLAS, MODU Code, HSLC Code): radar, AIS, GNSS, ECDIS, gyro, heading and bearing indicator, BNWAS, echo sounder, electronic plotting, speed and distance measuring device, tracking aid, rate of turn indicator and transmitting heading device. |
| Internal communication                                                                                                                                                                                                           | Internal communication systems required by applicable statutory regulations. E.g. public announcement, general alarm, two-way voice communication.                                                                                                                                                                                     |
| External communication                                                                                                                                                                                                           | External communication systems required by applicable statutory regulations. E.g. Global maritime distress and safety system.                                                                                                                                                                                                          |
| <sup>1)</sup> Systems providing local, remote or automatic control as well as required monitoring, alarm, and safety functions shall be included in the SuC.<br>Integrated control systems shall be included in the default SuC. |                                                                                                                                                                                                                                                                                                                                        |



# DNV is well positioned with more than 250 vessels and systems contracted for voluntary approval so far

Strong demand from owners and vendors with >250 voluntary approvals contracted

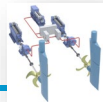
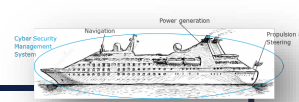
Implementing class approval in all areas with training engineers and surveyors

Cyber secure Essential class notation well aligned with IACS UR

We work with key maritime suppliers and yards to ensure their designs are compliant

- Voluntary DNV rules since 2018 in line with IACS UR and IEC
- Voluntary class notation has allowed to build competence internally and externally
- July 2023 rules introduced system certification in Cyber secure rules
- 2024 extraordinary rule update to make it mandatory from July 2024

# DNV Cyber secure class levels for ships & offshore units



**Cyber Secure Class Notation**  
(DNV-RU-SHIPS Pt.6 Ch.5 Sec.21)

**Cyber Secure Type Approval**  
(DNV-CP-0231)

Different levels

IACS UR  
2024

Entry-level for all merchant vessels



Cyber secure <no qualifier>

Security Profile 0

Newbuildings & high-end vessels



Cyber secure Essential

Security Profile 1

Advanced newbuildings with very high security

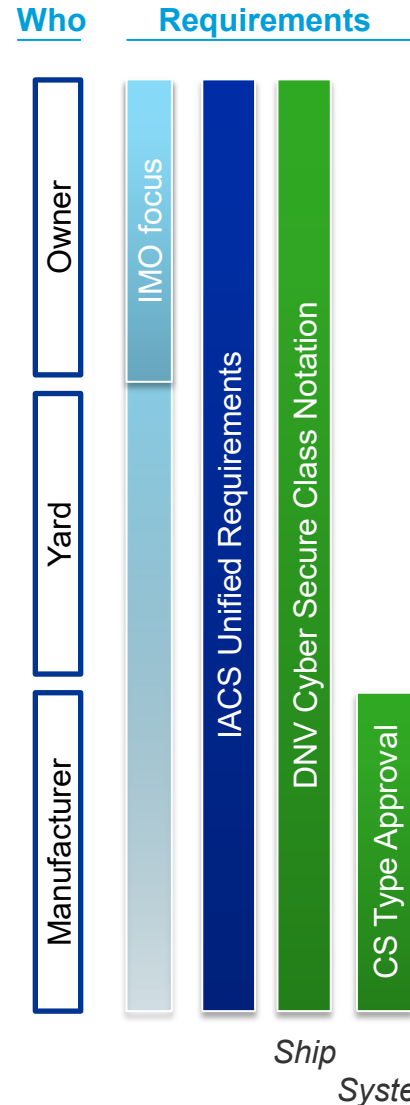
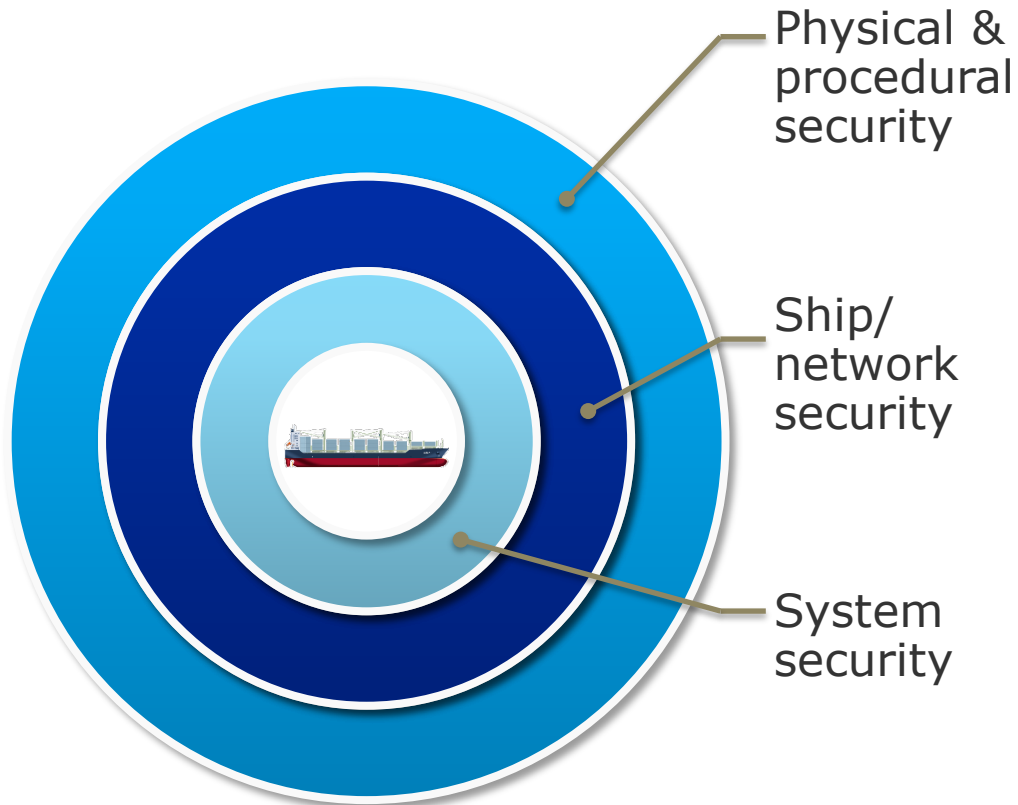


Cyber secure Advanced

Security Profile 3

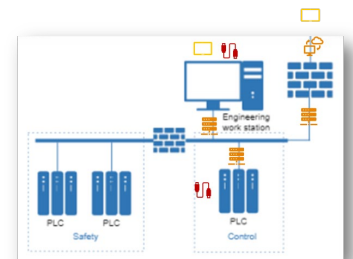
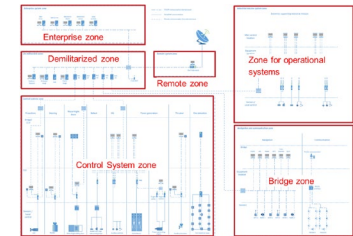
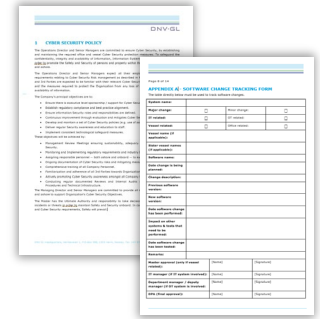
# Effective cyber security barriers using defence in depth concept

## Defence in depth model



## All layers support cyber resilience

- **Physical & procedural barriers** protect vessel systems using physical access and procedures (patching, access cards, locked cabinets, maintenance scans, “human firewall”, ...)
- **Ship barriers** protect connection between zones & systems with remote access (VPN, DMZ, ...), segregation (firewalls, data diodes, ...)
- **System barriers** protect the individual system with barriers such as encryption, user control and authentication, removable devices, event logging, backup and recovery, etc.



# 1<sup>st</sup> layer – Procedures and Management system

## Management system

Based on requirements from IEC 62443-2-1, including e.g.:

- Organizing for security
- Staff training and security awareness
- System inventory, major devices, segmentation and physical location
- Physical and environmental security
- Network segmentation
- Access control: Account administration and Authentication
- Incident planning and response
- Business continuity plan

1 CYBER SECURITY POLICY

The Operations Director and Senior Managers are committed to and maintaining the required office and vessel Cyber Security confidentiality, integrity and availability of Information, Informal grdec iq promote the Safety and Security of persons and proper and ashore.

The Operations Director and Senior Managers expect all requirements relating to Cyber Security Risk management as de and 3rd Parties are expected to be familiar with their relevant C and the measures required to protect the Organization from a availability of information.

The Company's principal objectives are to:

- Ensure there is executive level sponsorship / support for
- Establish regulatory compliance and best practice alignm
- Ensure information Security roles and responsibilities are
- Continuous improvement through evaluation and mitigat
- Develop and maintain a set of Cyber Security policies (e.)
- Deliver regular Security awareness and education to staff
- Implement consistent technological safeguard measures.

These objectives will be achieved by:

- Management Review Meetings ensuring sustainability, Security.
- Monitoring and Implementing regulatory requirements ar
- Assigning responsible personnel — both ashore and onbo
- Ongoing documentation of Cyber Security risks and mitig
- Comprehensive training of all Company Personnel.
- Familiarization and adherence of all 3rd Parties towards C
- Actively promoting Cyber Security awareness amongst al
- Conducting regular documented Reviews and Intern Procedures and Technical Infrastructure.

The Managing Director and Senior Managers are committed to j and ashore to support Organization's Cyber Security Objectives.

The Master has the Ultimate Authority and responsibility to t incidents or threats in, grdec iq maintain Safety and Security onb and Cyber Security requirements, Safety will prevail.]

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APPENDIX A| - SOFTWARE CHANGE TRACKING FORM

The table directly below must be used to track software changes.

|                                                                 |                          |                 |                          |
|-----------------------------------------------------------------|--------------------------|-----------------|--------------------------|
| System name:                                                    |                          |                 |                          |
| Major change:                                                   | <input type="checkbox"/> | Minor change:   | <input type="checkbox"/> |
| IT related:                                                     | <input type="checkbox"/> | OT related:     | <input type="checkbox"/> |
| Vessel related:                                                 | <input type="checkbox"/> | Office related: | <input type="checkbox"/> |
| Vessel name (if applicable):                                    |                          |                 |                          |
| Sister vessel names (if applicable):                            |                          |                 |                          |
| Software name:                                                  |                          |                 |                          |
| Date change is being planned:                                   |                          |                 |                          |
| Change description:                                             |                          |                 |                          |
| Previous software version:                                      |                          |                 |                          |
| New software version:                                           |                          |                 |                          |
| Date software change has been performed:                        |                          |                 |                          |
| Impact on other systems & tests that need to be performed:      |                          |                 |                          |
| Date software change has been tested:                           |                          |                 |                          |
| Remarks:                                                        |                          |                 |                          |
| Master approval (only if vessel related):                       | [Name]                   | [Signature]     |                          |
| IT manager (if IT system involved):                             | [Name]                   | [Signature]     |                          |
| Department manager / deputy manager (if OT system is involved): | [Name]                   | [Signature]     |                          |
| DPA (final approval):                                           | [Name]                   | [Signature]     |                          |

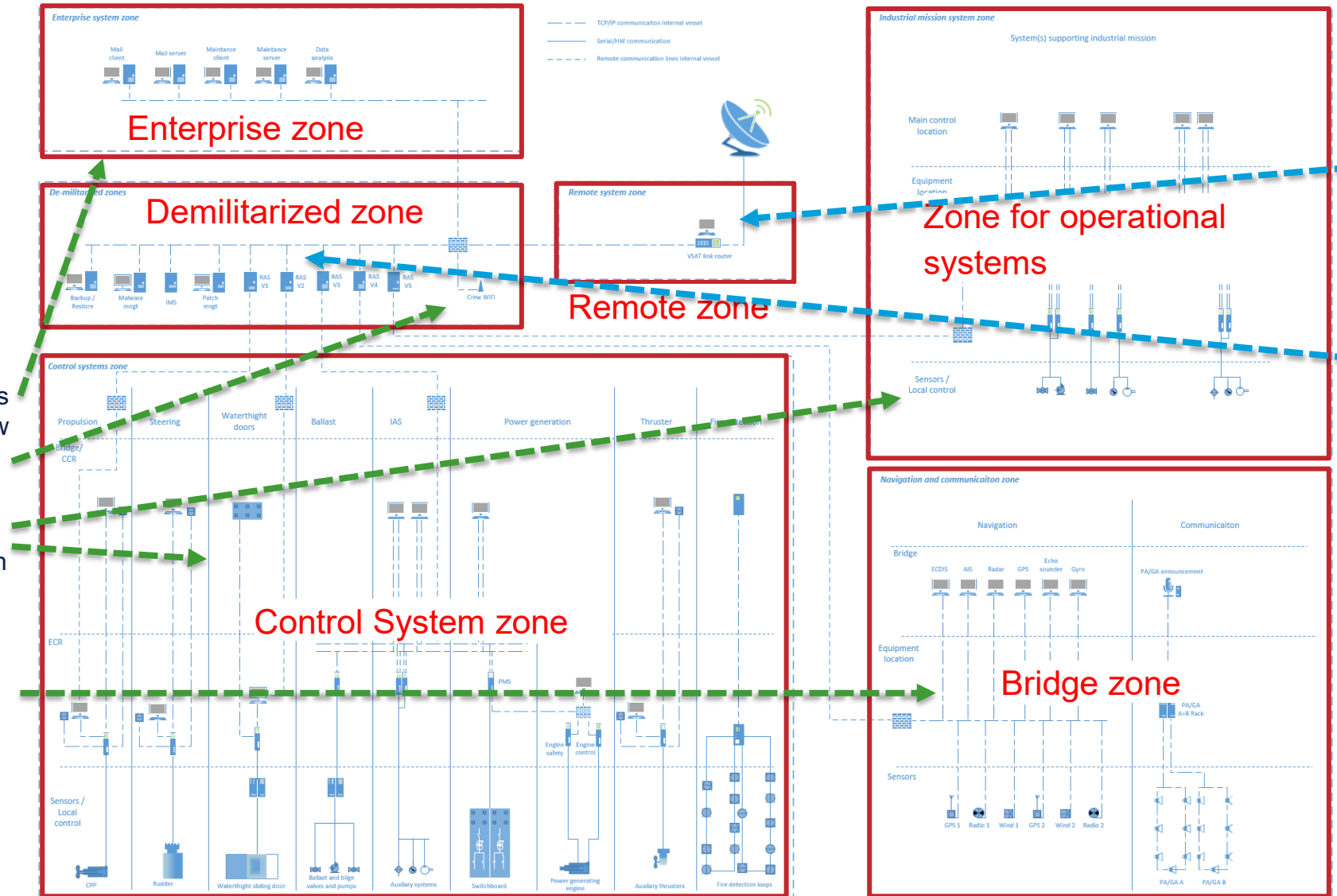


# 2<sup>nd</sup> layer - Vessel and network security

## Firewall & segregation

Key to design vessel where systems grouped in zones according to risk and attack surfaces, e.g.

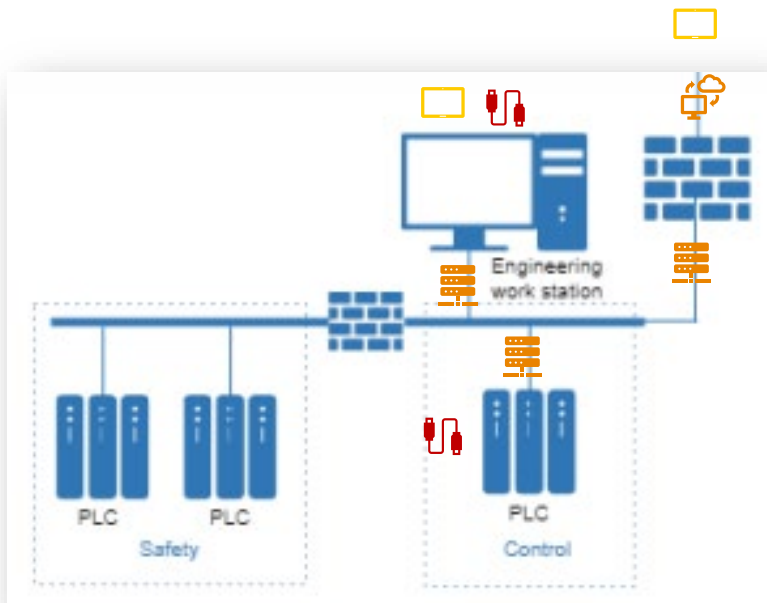
- Corporate IT networks segregated from Crew IT /entertainment
- Control and drilling systems with strong barriers & segregation
- Navigation & communication systems on separate network from both control and IT



## Remote access

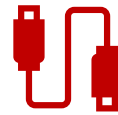
- Any remote access into vessel for data for business use, e.g. energy mgmt./ commercial use ensure read only
- Any remote maintenance for control systems onboard vessel strong security barriers with multi-factor authentication and encryption (as internet banking)
- These connections always controlled by crew onboard

# 3<sup>rd</sup> layer – Protecting the automation and navigation systems



Detect &  
respond

Secure  
development



- **Removable devices** - common attack surface for control systems. Necessary and easy to use tool for many critical features of the system such as exporting logs and data, updating the system and doing back-up and recovery



- **Network interfaces to other and remote systems** - attack surface to control systems, especially as more systems are based on common network technology and have remote access



- **Human machine interfaces** - attack surface typically requires physical access to the main installation or a remote panel

# Defence in depth – linking DNV rules and IACS UR

## Management of cyber security - E26 - Shipowner

- Scope of applicability
- Cyber risk management
- Change management
- Patch management
- Classification of information
- Access control
- Management of firewalls
- Malware protection
- Remote access
- Use of mobile devices
- Detection of anomalies
- Verification of security functions
- Incident response
- Incident recovery
- Backup and restore

## Security zones and conduits - E26 - Shipyard

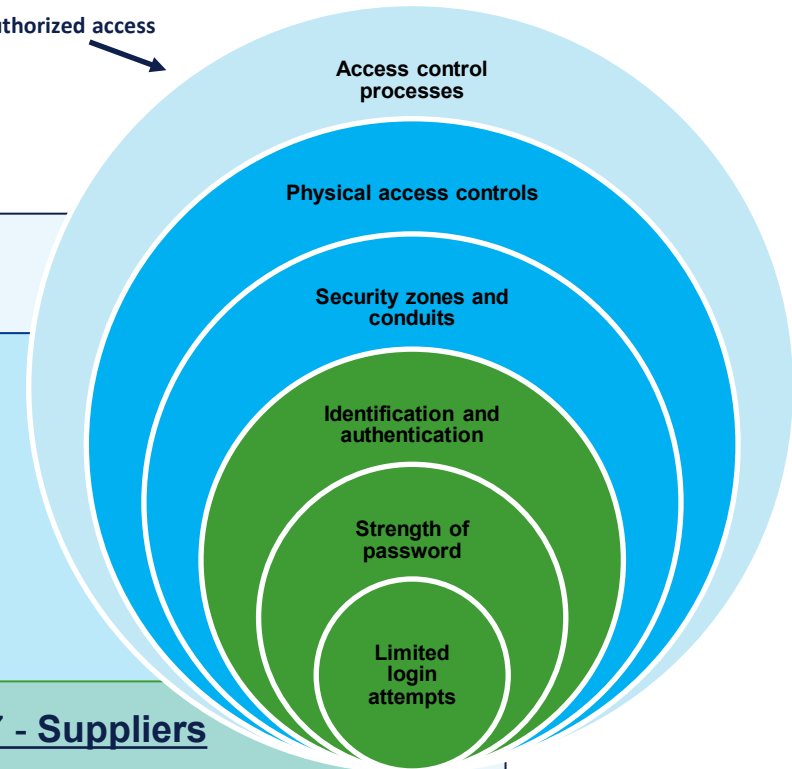
- Security zones and conduits
- Network segmentation
- Control of zone boundaries/conduits
- Implementation, integration, physical security

## Certification of system security capabilities - E27 - Suppliers

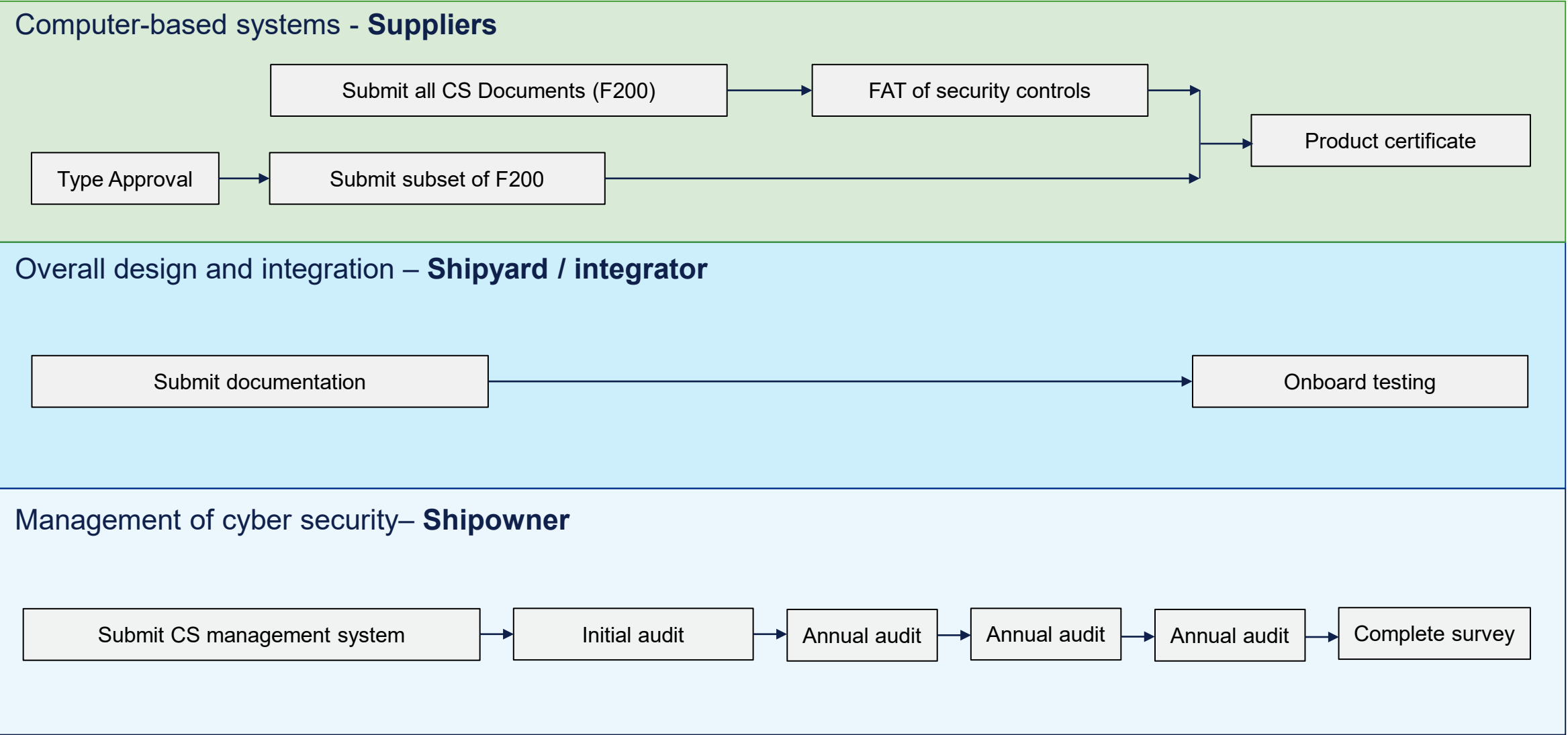
- 30 + 11 security capabilities based on IEC 62443-3-3
- Implementation/configuration of security capabilities
- Hardening
- SDL process



Ex. threat: Unauthorized access

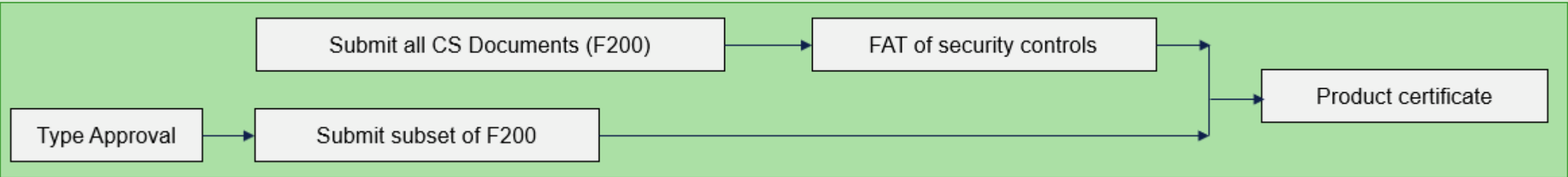


# Overview of the classification process



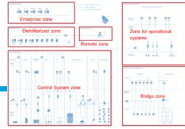


# Classification process - Suppliers



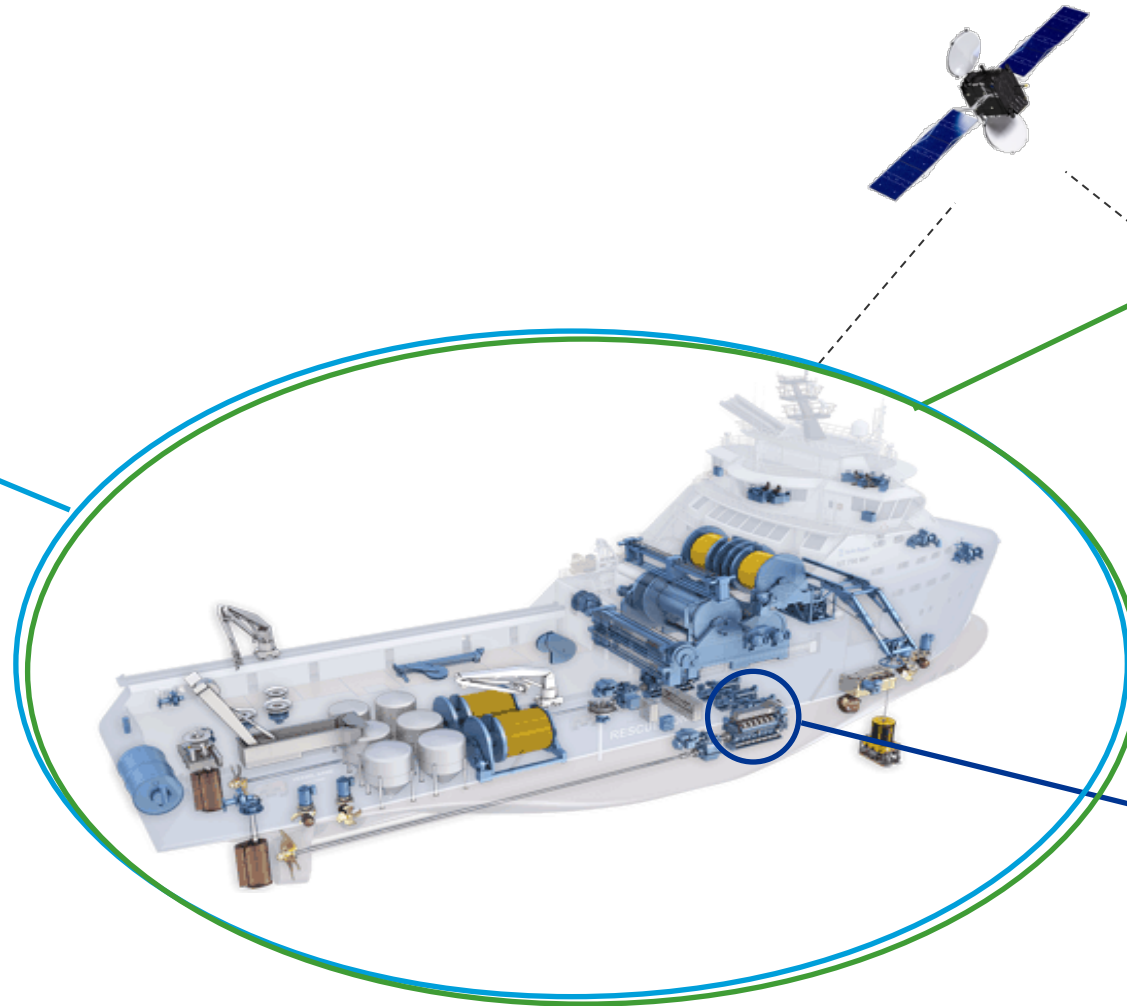
| Object                                                                 | Documentation type                              | Additional description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Info |
|------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| System in SuC                                                          | F200 - Cyber security documentation for systems | For systems with type approved security capabilities, F200 shall include the following documents:<br>— F030 - System topology<br>— F071 - Inventory list<br>— F262 - Test report or declaration of hardening and security configuration                                                                                                                                                                                                                                                                                                                                                                        | AP   |
|                                                                        |                                                 | For systems without type approved security capabilities, F200 shall include the following documents:<br>— F030 - System topology<br>— F071 - Inventory list<br>— F021 - Description of security capabilities<br>— F252 - Procedure for test of security capabilities<br>— F259 - Description of secure development lifecycle processes<br>— F141 - Guidelines for hardening and configuration of security capabilities<br>— F142 - Information supporting incident response and recovery<br>— F253 - Procedure for verification of security functions on board<br>— I320 - Software change handling procedure. | AP   |
| See requirements for document content in <a href="#">DNV-CG-0550</a> . |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |

# All parts of DNV impacted by new cyber secure rules



## Newbuilding

- NB design approval
- Site survey

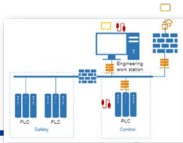


## Fleet in Service

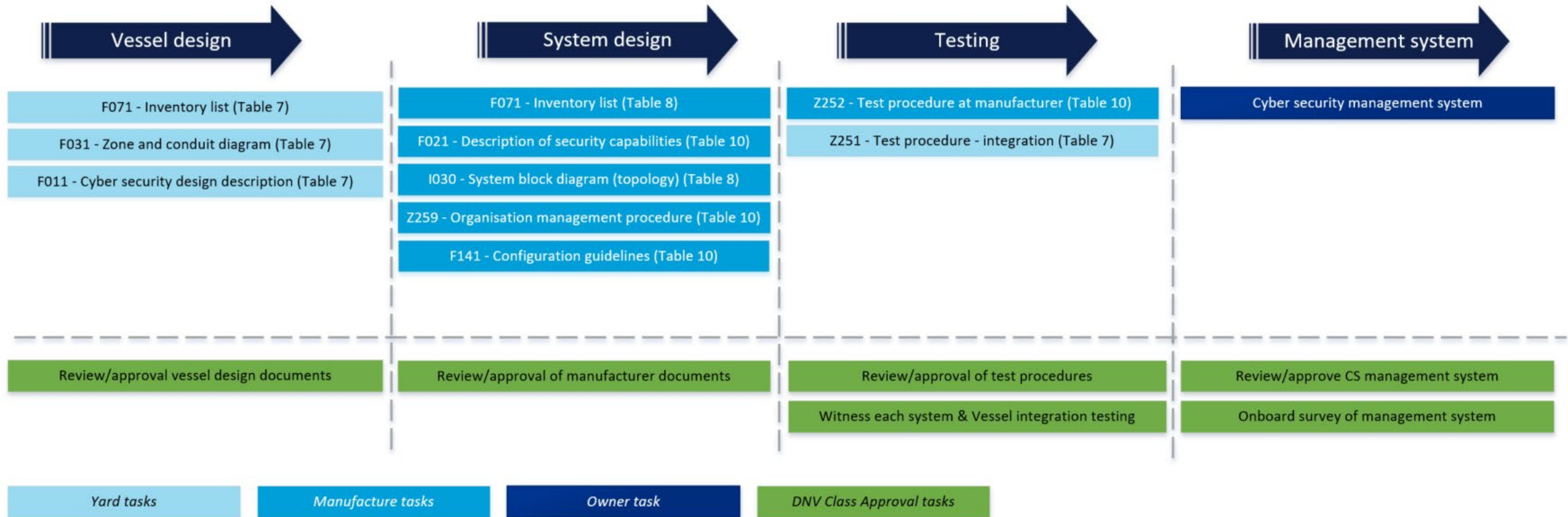
- CSMS plan approval
- Initial survey
- Annual survey
- Complete survey

## System certification

- Product certification with plan approval and survey
- Type approval for reduced product certification



# Process & documents - Cyber secure Class notation



# Required integrator actions - yards & designers

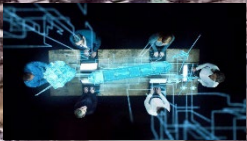


# Newbuilding related cyber security class activities

Shipyard



IACS E26



**NB CS design approval** including overall vessel design and integration (network, remote access, ...) by DNV Approval centres

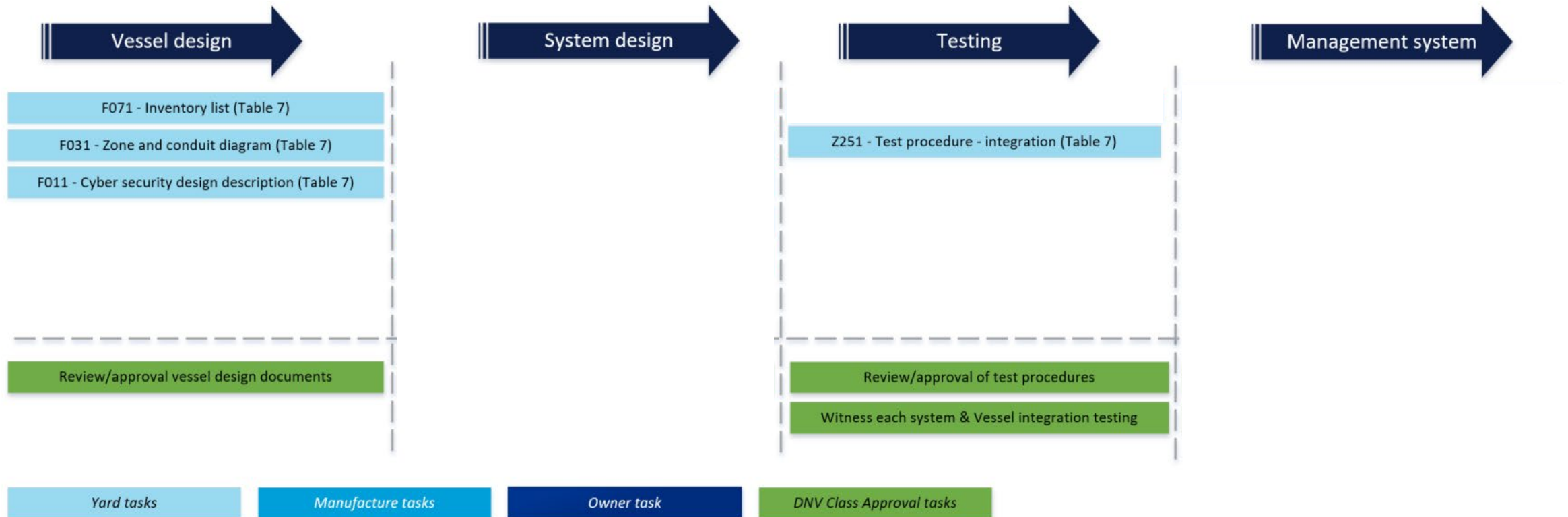


**NB CS Site survey/test witnessing** including integration test and verification of systems in scope to be done by each DNV site office



# Process & documents - Cyber secure Class notation

## Yard/designer delivery:



# Initial system selection

## - risk assessment systems under consideration

### 2.2 Exemption based on negligible risk

A risk assessment may be carried out for the purpose of exempting a computer-based system from the security requirements in [4]. The objective is to determine if the system may be considered to represent negligible cyber risk. The risk assessment shall at least consider the criteria listed below:

- The system shall be isolated and not have network communication with other systems (including no remote access solution).
- The system shall be located in area(s) to which physical access is restricted and controlled.
- The system shall not have accessible physical interface ports. Unused interfaces shall be logically disabled. It shall not be possible to mount external devices to the system.
- The system shall not be an integrated control system serving multiple vessel functions, see [Pt.4 Ch.9 Sec.2 \[1.3\]](#).

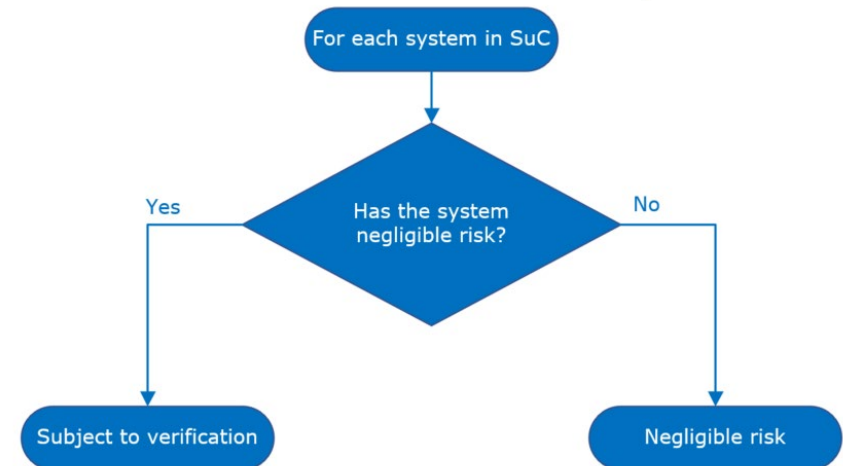
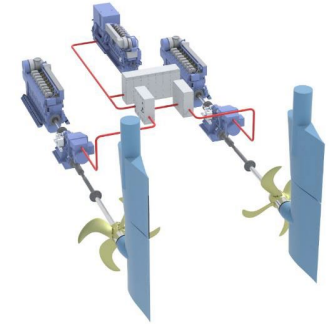
The following additional criteria should also be considered in the risk assessment:

- The system should not be needed for propulsion and steering, and should not include required safety functions.
- The system should not be susceptible by cyber incidents vectored by other systems or network devices, nor propagate the effect of a cyber incident to other systems or network devices.
- Known vulnerabilities, threats, attack surface, and potential impacts deriving from a cyber incident should be considered

A risk assessment may also be carried out for the purpose of exempting a component in the system from all or some security requirements in [4]. In such case the risk assessment should consider each required security capability and propose compensating countermeasures for applicable security capabilities that are not implemented in the component.

The risk assessment is subject to approval by the Society as follows:

- For a system to be exempted from SuC, the risk assessment shall be submitted by the integrator as part of document F011, see [Table 7](#).
- For a component in the system to be exempted from all or some security requirements, the risk assessment shall be submitted by the manufacturer as part of document F021, see [Table 10](#).



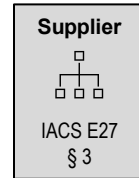
# Asset inventory

You can't protect your assets unless you know what your assets are

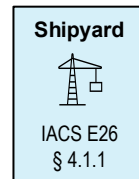
## Asset inventory

[illegible]

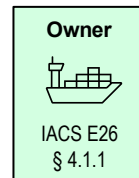
*In collaboration with vendors the following information is documented : hardware model(s) and version, connection points (USB, Ethernet, etc.), address lists, e.g. IP address lists, firmware and software application versions, configuration settings, patch levels, malware protection, hardening, internal and external interfaces, etc.*



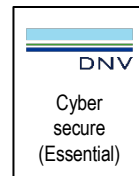
Deliver asset inventory as part of system documentation



Deliver the ship with a complete and updated inventory of hardware and software for the systems in scope.



## Keep asset inventory updated

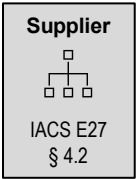
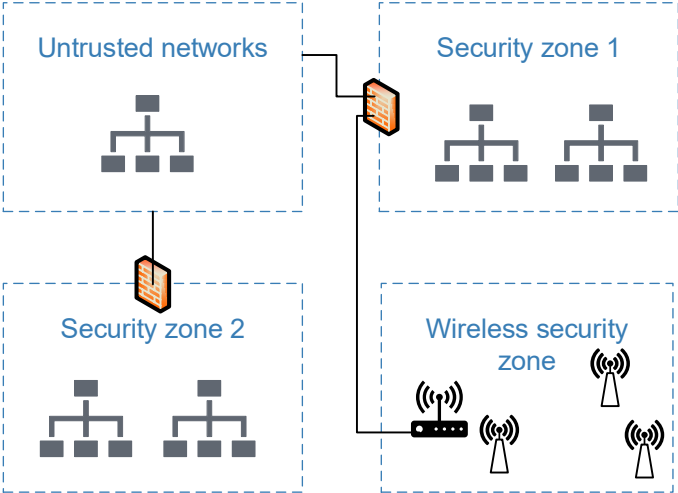


Aligned

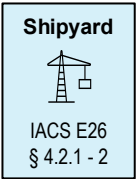
# Security zones

Systems to be grouped in security zones, and zone boundaries (conduits) to be protected.

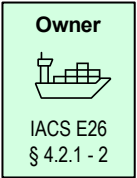
Safety, navigation and wireless systems allocated to separate security zones



Coordinate any communication through conduits with shipyard, note E27 §4.2



Design security zones, control communication through conduits



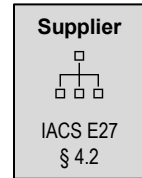
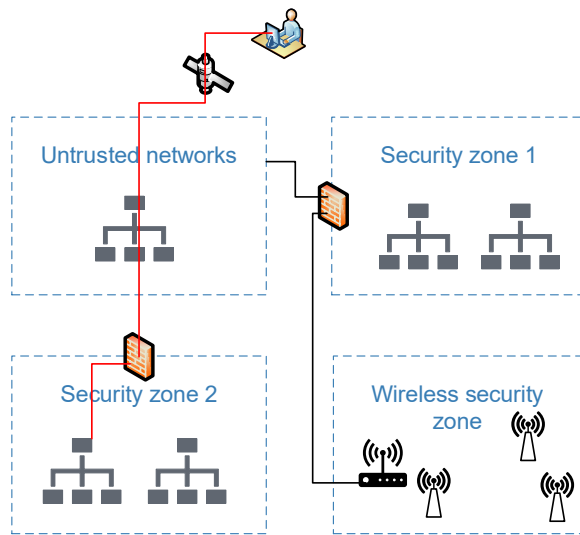
Manage the zones and conduits in operation



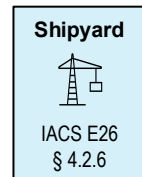
Aligned

# Remote access / untrusted networks

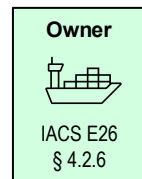
Prevent cyber incidents originating from external/untrusted networks



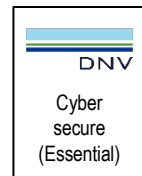
- ✓ Multifactor authentication (SR 1.1 RE1)
- ✓ Endpoint authentication (SR 1.2)
- ✓ Limit consecutive login attempts (SR 1.11)
- ✓ Use notification (SR 1.12)
- ✓ Monitor and control remote access (SR 1.13)
- ✓ Explicit access request approval (SR 1.13 RE1)
- ✓ Remote session termination (SR 2.6)
- ✓ Cryptographic integrity protection (SR 3.1 RE1)
- ✓ Session integrity (SR 3.8 / SR 3.8 RE1)



Shipyard to deliver user manual for remote access, ensure the systems are not exposed to external networks, ensure systems meet E27 § 4.2



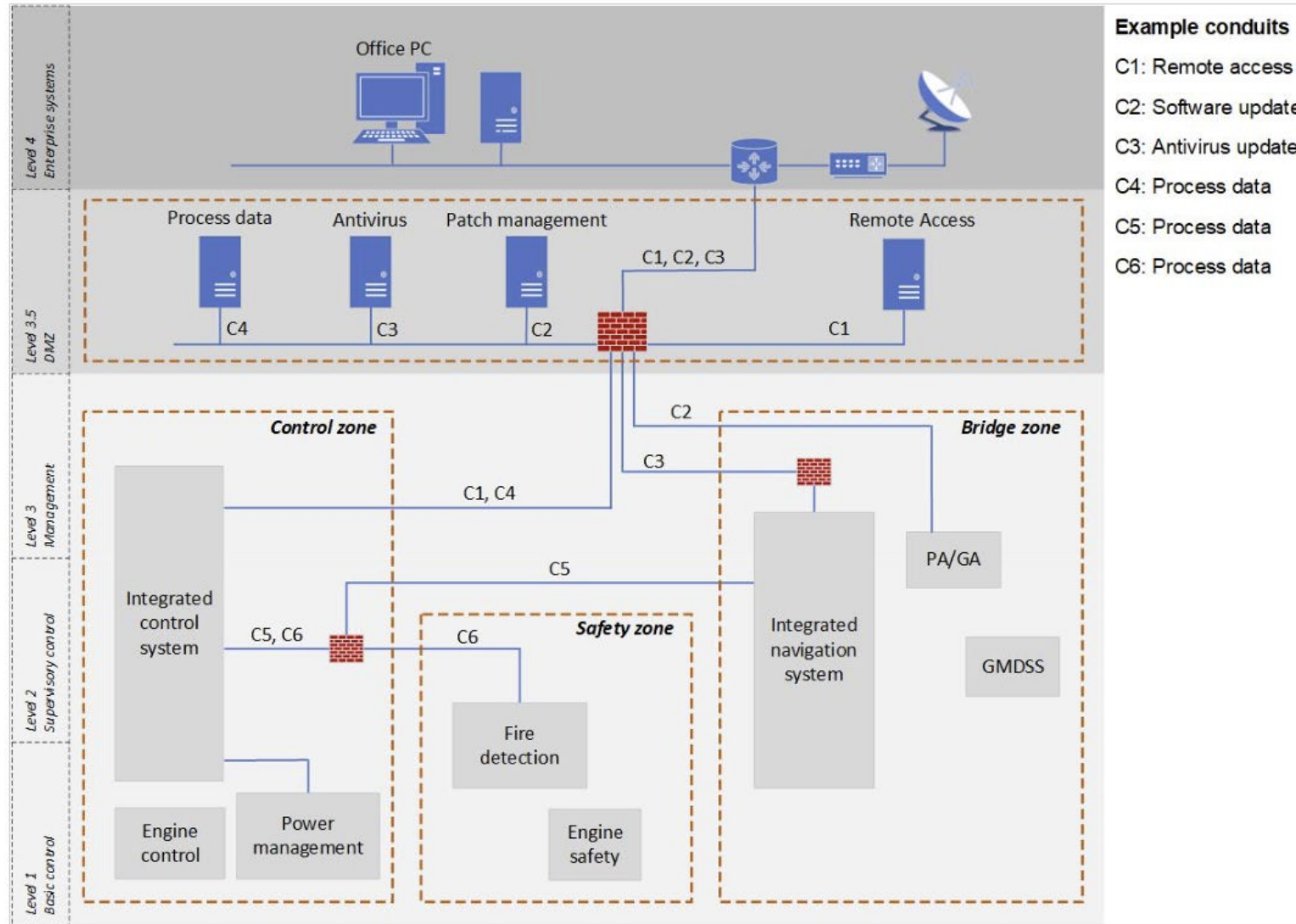
Manage communication between security zones and untrusted networks in operation



Aligned



# Zone and conduit drawing in DNV Cyber secure rules



# Cyber security design description

- Template available
- Description of each security zone
- Description of each conduit between zones and to off-ship systems
- A risk assessment for each system exempted from scope
- Overall vessel cyber security controls

|                                                                                                                                               |                                                              |
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| 8.1.5                                                                                                                                         | Bridge zone ..... 4                                          |
| 8.1.6                                                                                                                                         | Other zones ..... 4                                          |
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| 9.1.1                                                                                                                                         | Requirements to comply with ..... 5                          |
| 9.1.2                                                                                                                                         | Other requirement details ..... 5                            |
| 9.2                                                                                                                                           | Bridge zone ..... 5                                          |
| 9.2.1                                                                                                                                         | requirements to comply with ..... 5                          |
| 9.2.2                                                                                                                                         | Other requirements details ..... 5                           |
| 9.3                                                                                                                                           | Remote zone ..... 5                                          |
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| 9.3.2                                                                                                                                         | Other requirements details ..... 6                           |
| 9.4                                                                                                                                           | DMZ zone ..... 6                                             |
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| 9.4.2                                                                                                                                         | Other requirement details ..... 6                            |
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| 9.5.1                                                                                                                                         | Requirements to comply with ..... 6                          |
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| 9.6                                                                                                                                           | Other zones ..... 6                                          |
| 9.6.1                                                                                                                                         | Requirements to comply with ..... 6                          |
| 10                                                                                                                                            | FUNCTION COMPLIANCE FOR SAILING VESSELS (NOT FOR NB) ..... 7 |
| <br>DNV Headquarters, Veritasveien 1, P.O.Box 300, 1322 Høvik, Norway. Tel: +47 67 57 99 00. <a href="http://www.dnvgl.com">www.dnvgl.com</a> |                                                              |

# Required supplier actions - system manufacturer

# Certification of Materials and Components (CMC) related cyber security class activities

Supplier

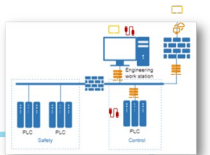
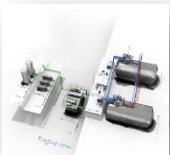


IACS E27

## Mandatory product certification



CS Type approval?



### Full Case-by-case w/o TA

All systems in scope and not deemed negligible risk (no network, SW, updates, access, ...) will require

- i. **CS system design approval** – DNV approval centre
- ii. **CS system survey** – by local DNV office

### Reduced Case-by-case with TA

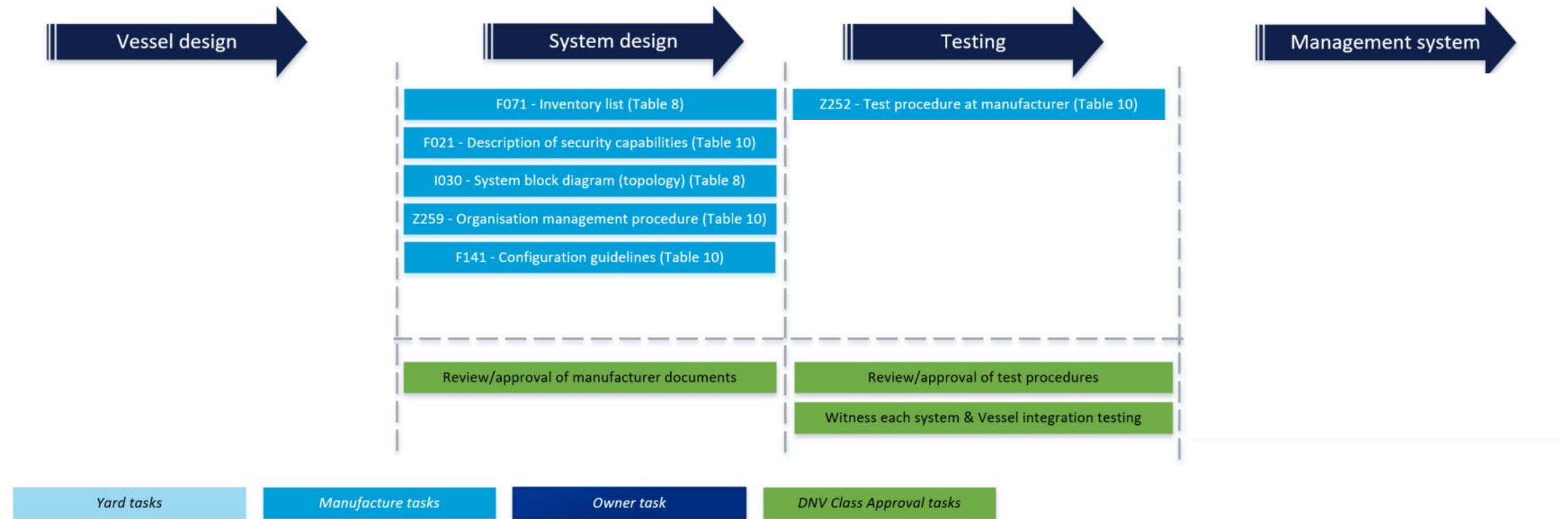
All systems in scope and not deemed negligible risk (no network, SW, updates, access, ...) will require

- i. **Reduced CS system design approval** – confirm applicability of TA/configuration/supplier test report
- ii. **CS system survey** – omitted



# Process & documents - Cyber secure Class notation

## System manufacturer:





# IACS UR and DNV rules - security capabilities

## Identification and authentication

- SR 1.1 Identification and authentication (#)
- SR 1.3 Account management (#)
- SR 1.4 Identifier management (#)
- SR 1.5 Authenticator management (#)
- SR 1.6 Wireless access management (#)
- SR 1.7 Strength of passwords
- SR 1.10 Authenticator feedback

*DNV SP1 and IACS UR E27*

## System integrity

- SR 3.1 Communication integrity (#)
- SR 3.2 Malicious code protection (#)
- SR 3.3 Security functionality verification (#)
- SR 3.6 Deterministic output (#)

*DNV SP1 and IACS UR E27*

## Resource availability

- SR 7.1 Denial of service protection (#)
- SR 7.2 Resource management (#)
- SR 7.3 Backup (#)
- SR 7.4 Restore (#)
- SR 7.5 Emergency power
- SR 7.6 Configuration guidelines
- SR 7.7 Least functionality (#)

*DNV SP1 and IACS UR E27*

## Secure Development Lifecycle (SDL)

- SM-8 Control of private keys
- SUM-2 Security update documentation
- SUM-3 Dependent comp. security update docum.
- SUM-4 Security update delivery
- SG-1 Product defence in depth
- SG-2 Defence in depth in the project
- SG-3 Hardening guidelines

*DNV SP1 and IACS UR E27*

## Use control

- SR 2.1 Authorization enforcement (#)
- SR 2.2 Wireless use control (#)
- SR 2.3 Portable and mobile devices (#)
- SR 2.4 Mobile code
- SR 2.5 Session lock (#)
- SR 2.8 Auditable events (#)
- SR 2.9 Audit storage capacity
- SR 2.10 Audit processing failures
- SR 2.11 Timestamps (SL2)

*DNV SP1 and IACS UR E27*

## Restricted data flow (UR E26)

- SR 5.1 Network segmentation (\*)
- SR 5.1 RE1 Physical network segmentation (\*)
- SR 5.2 Zone boundary protection (\*)
- SR 5.2 RE1 Deny by default (\*)
- SR 5.3 General purpose P2P
- SR 5.4 Application partitioning

*DNV SP1 and IACS UR E26 (\*)*

## Data confidentiality

- SR 4.1 Confidentiality of "sensitive" information (#)
- SR 4.3 Use of cryptography

*DNV SP1 and IACS UR E27*

## Interfaces with other/untrusted networks

- SR 1.1 RE2 Multifactor authentication (SL3) (#)
- SR 1.2 Endpoint authentication ((SL2) #)
- SR 1.11 Unsuccessful login attempts (SL1)
- SR 1.12 System use notification (SL1)
- SR 1.13 Access via untrusted networks (SL1) (#)
- SR 1.13 RE1 Explicit access approval (SL2) (#)
- SR 2.6 Remote session termination (SL2) (#)
- SR 3.1 RE1 Cryptographic integrity prot. (SL3) (#)
- SR 3.5 Input validation (SL1)
- SR 3.8 Session integrity (SL2)
- SR 3.8 RE1 Invalidation of session ID (SL3)

*DNV SP1 and IACS UR E27*

## Timely response to events

- SR 6.1 Audit log accessibility (#)

*DNV SP1 and IACS UR E27*

## Product certification / Type approval



(#) also required by UR E26

# Activities, suppliers

## Preparations (now)

Determine if system is in scope

Implement required security capabilities

Establish Secure Development Lifecycle processes

Evaluate feasibility of Type approval

## For each delivery - if system is type approved

Submit reduced set of documents to class society

- F030 - System topology
- F071 - Inventory list
- F262 - Test report or declaration of hardening and security configuration

Obtain Product certificate

## For each delivery - if system is not type approved

Submit complete set of documents to class society

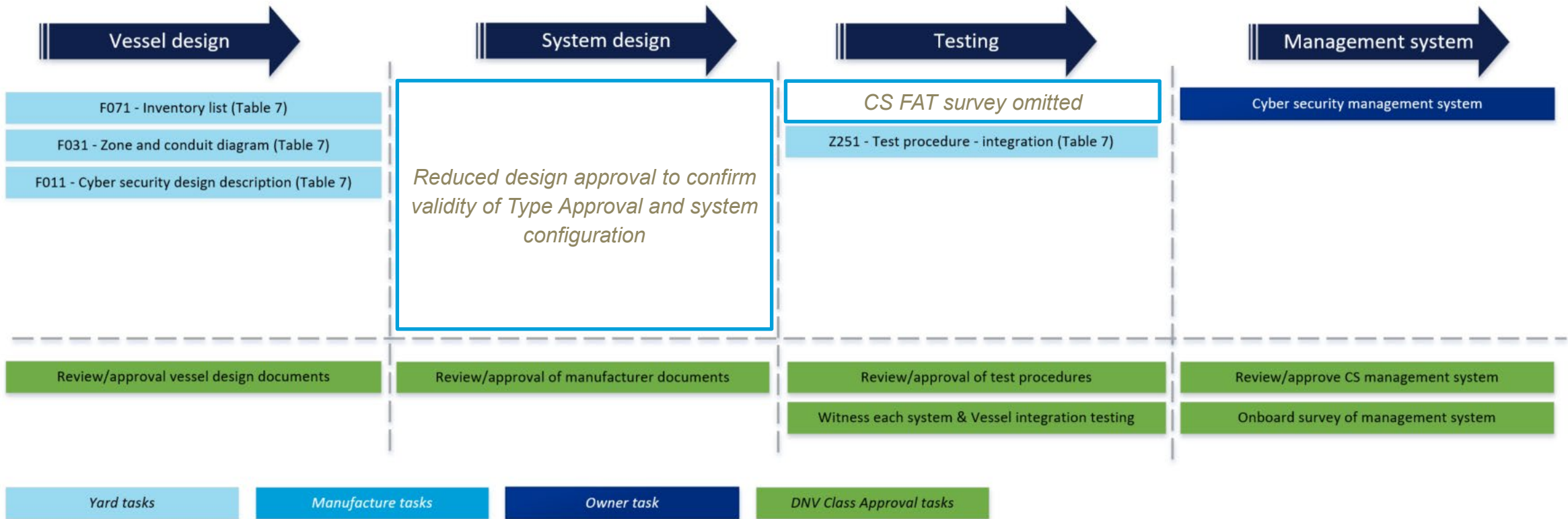
- F030 - System topology
- F071 - Inventory list
- F021 - Description of security capabilities
- F252 - Procedure for test of security capabilities
- F259 - Description of secure development lifecycle processes
- F141 - Guidelines for hardening and configuration of security capabilities
- F142 - Information supporting incident response and recovery
- F253 - Procedure for verification of security functions on board
- I320 - Software change handling procedure.

Factory acceptance test

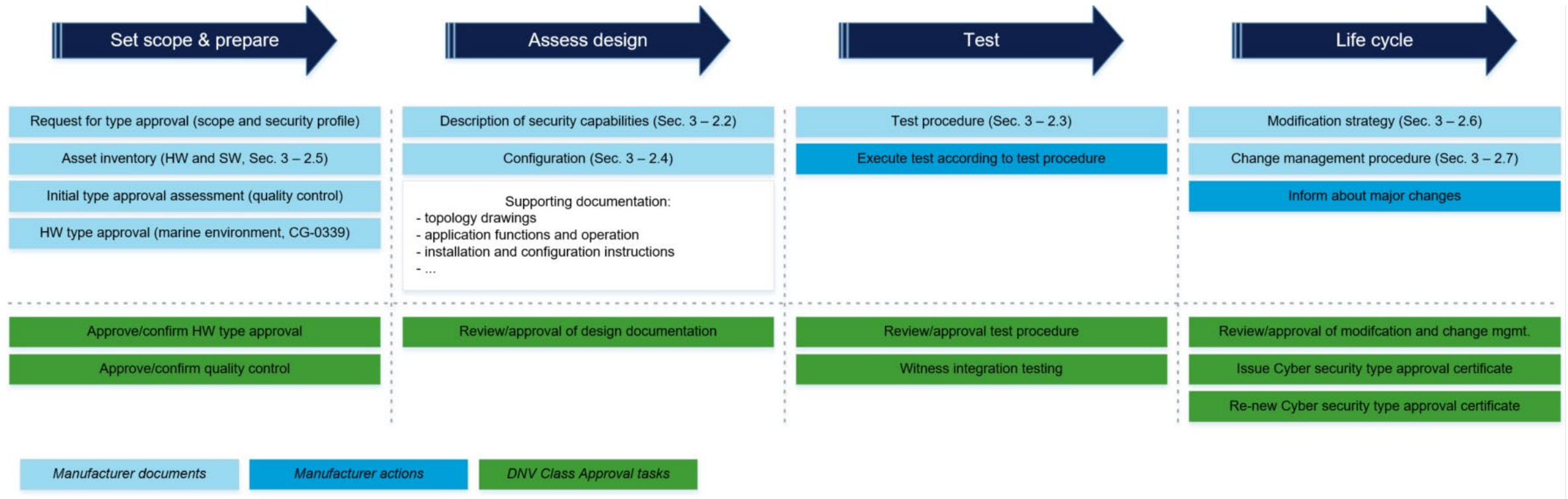
Obtain Product certificate

# Process & documents - Cyber secure Class notation

## For type approved systems:



# Type approval confirms design, implementation, testing & life cycle of cyber security barriers



Required actions:

- ship owner & manager



# Summary – DNV Cyber secure and IACS UR E26/27

# Fleet in Service related Cyber security class activities



- **Approval of Cyber security management system** by DNV
- **Initial survey by DNV surveyor** to check in depth implementation of cyber security procedures as well as covering compensating measures to address design weaknesses
- **Annual survey by DNV surveyor** verify that procedures are being followed
- **Complete survey** ensures that both procedural and technical barriers are still in place

**Initial survey**  
Initial survey check CS procedures in-depth

**Annual Survey**

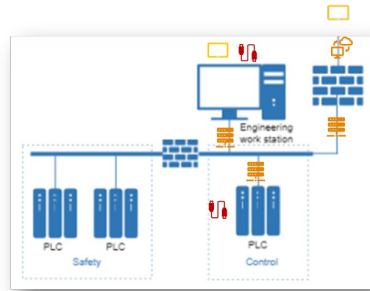
- Focus on procedures
- Optionally as remote survey

**Complete survey**

- Re-testing of cyber security defences witnessed by DNV



# Key activities



Supplier



IACS E27

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| 8.1.2    | Bridge zone                                          |
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| 8.1.4    | Control system zone                                  |
| 8.1.5    | Bridge zone                                          |
| 8.1.6    | Other zones                                          |
| 8.2      | Network infrastructure and ports                     |
| 8.2.1    | Additional network ports                             |
| 8.2.2    | Temporary device ports                               |
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| 9.4      | Bridge zone                                          |
| 9.4.1    | Requirements to comply with                          |
| 9.4.2    | Other requirements details                           |
| 9.5      | Bridge zone                                          |
| 9.5.1    | Requirements to comply with                          |
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| 10       | FUNCTION COMPLIANCE FOR SAILING VESSELS (NOT FOR ME) |

Shipyard



IACS E26

| 1 CYBER SECURITY POLICY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| The Operations Director and Senior Managers are committed to ensure Cyber Security, by establishing and maintaining the required cyber and vessel Cyber Security protection measures. To safeguard the confidentiality, integrity and availability of information, information systems and IT / OT equipment. DNV GL provides the Safety and Security of persons and property within the Organization both onboard and ashore.                                                                                                                                                                                                                                                                                                                                                                                                       |  |
| The Operations Director and Senior Managers expect all their employees to comply with the requirements relating to Cyber Security Risk management as described in MSC-A28 (88). All employees and all Parties are expected to be familiar with their relevant Cyber Security duties and responsibilities and the measures required to protect the Organization from any loss of confidentiality, integrity and availability of information.                                                                                                                                                                                                                                                                                                                                                                                          |  |
| The Company's principal objectives are to:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| <ul style="list-style-type: none"> <li>• Ensure there is adequate asset identification / support for Cyber Security.</li> <li>• Ensure regulatory compliance and best practice engagement.</li> <li>• Ensure information Security roles and responsibilities are defined.</li> <li>• Continuous improvement through evaluation and mitigation Cyber Security risks.</li> <li>• Develop and maintain a set of Cyber Security policies (e.g. use of assets).</li> <li>• Develop regular Security awareness and education to staff.</li> <li>• Implement consistent technological safeguard measures.</li> </ul>                                                                                                                                                                                                                        |  |
| These objectives will be achieved by:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| <ul style="list-style-type: none"> <li>• Management Review Meetings ensuring sustainability, adequacy and effectiveness of Cyber Security.</li> <li>• Monitoring and implementing regulatory requirements and industry best practices.</li> <li>• Assigning responsible personnel – both onboard and ashore – to handle relevant tasks.</li> <li>• Ongoing documentation of Cyber Security risks and mitigating measures.</li> <li>• Comprehensive training of all Company Personnel.</li> <li>• Familiarization and adherence of all 3rd Parties towards Organization's Cyber Security Policies.</li> <li>• Active promoting Cyber Security awareness onboard at Company Personnel.</li> <li>• Conducting regular documented Reviews and Internal Audit of Cyber Security Policies, Procedures and Operational Controls.</li> </ul> |  |
| The Managing Director and Senior Managers are committed to provide all necessary resources onboard and ashore to support Organization's Cyber Security Objectives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
| The Master has the ultimate Authority and responsibility to take decisions regarding Cyber Security incidents or threats (e.g. MSC-A28, Maintain Safety and Security onboard). In case of conflict between Safety and Cyber Security requirements, Safety prevails.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |

Owner



IACS E26

- Determine if system is in scope and not negligible risk
- Design security capabilities as per UR E27
- Establish processes as per UR E27
- Evaluate feasibility of Type approval vs. Case-by-case certification
- Configuration and hardening
- Obtain compliance document for each delivery

- Evaluate exemption of systems based on negligible risk
- Acquire systems compliant with E27
- Design security zones, conduits, physical controls, countermeasures
- Ensure implementation and integration as per UR E26
- Produce and submit documentation
- Demonstrate/test security controls in commissioning

- Document and submit security management processes
- Implement security management on board
- Demonstrate management processes in annual surveys
- Demonstrate/test security controls in complete survey

# What to do now?



## Overall

- **Mandatory IACS Cyber security** requirements (UR E26/E27) for newbuild contracts from 1<sup>st</sup> of July 24'
- DNV Cyber secure Essential (SP1) class notation **fully aligned** with IACS UR
- **DNV well positioned** with >250 vessels / systems contracted for Cyber secure



## Owners and Ship managers

- New vessels contracted from 1<sup>st</sup> of July **require DNV Cyber secure Essential**
- **Voluntary for existing vessels** and newbuildings before this date
- **Optional lower-level** Cyber secure notation for existing vessels with lower risk



## Yards and designers

- Prepare **Cyber secure compliant design**
- Now consider voluntary class notation to **build competence**
- **Check if system suppliers are compliant** as they class certification

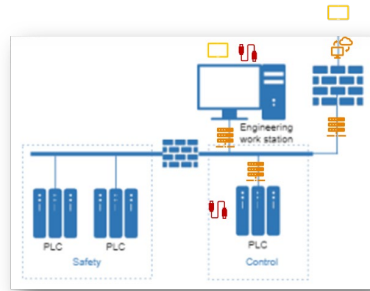


## Suppliers

- Prepare for **delivering compliant systems product certified** towards requirements
- **Consider CS Type approval** of systems to reduce case-by-case product certification effort

# Technical Cyber security requirements

# Key activities



Supplier



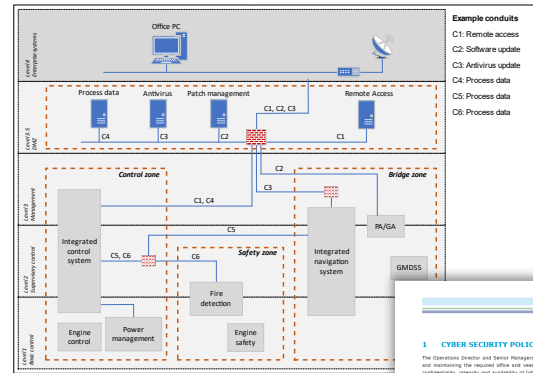
IACS E27

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| 9.4      | CNC zone                                             |
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Shipyard



IACS E26



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| The Managing Director and Senior Managers are committed to provide all necessary resources onboard and ashore to support Organisation's Cyber Security Objectives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| The Master has the ultimate authority and responsibility to take decisions regarding Cyber Security incidents or threats (e.g. MSC-A28, Maintain Safety and Security onboard). In case of conflict between Safety and Cyber Security requirements, Safety prevails.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |

Owner



IACS E26

- Determine if system is in scope and not negligible risk
- Design security capabilities as per UR E27
- Establish processes as per UR E27
- Evaluate feasibility of Type approval vs. Case-by-case certification
- Configuration and hardening
- Obtain compliance document for each delivery

- Evaluate exemption of systems based on negligible risk
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- Design security zones, conduits, physical controls, countermeasures
- Ensure implementation and integration as per UR E26
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- Document and submit security management processes
- Implement security management on board
- Demonstrate management processes in annual surveys
- Demonstrate/test security controls in complete survey



# Systems in scope for the IACS UR on cyber security

**Table 12 Default SuC**

| <i>Function</i>                                                                                                                                                                                                                  | <i>Examples of systems <sup>1)</sup></i>                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Propulsion                                                                                                                                                                                                                       | Systems needed for propulsion of the vessel. E.g. propeller pitch, engine speed, thruster, electrical motors and drives.                                                                                                                                                                                                               |
| Steering                                                                                                                                                                                                                         | Systems needed for steering of the vessel. E.g. rudder, azimuth thruster, electrical motors and drives.                                                                                                                                                                                                                                |
| Watertight integrity                                                                                                                                                                                                             | Systems needed to meet requirements for internal and external watertight integrity. E.g. internal watertight doors, shell doors, flooding detection.                                                                                                                                                                                   |
| Bilge, ballast                                                                                                                                                                                                                   | Systems needed to meet requirements for bilge drainage, and ballasting. E.g. valves, pumps, level measurement, loading computer                                                                                                                                                                                                        |
| Anchoring and mooring                                                                                                                                                                                                            | System(s) needed to meet requirements for anchoring and mooring in <a href="#">Pt.3 Ch.11 Sec.1</a> .                                                                                                                                                                                                                                  |
| Electrical power                                                                                                                                                                                                                 | Systems needed for generation and distribution of electrical power. E.g. main and auxiliary engine, generator, switchboard, energy storage system, power management.                                                                                                                                                                   |
| Auxiliary systems                                                                                                                                                                                                                | Systems needed to serve required auxiliary functions. E.g. cooling, fuel, lubrication, compressed air, ventilation.                                                                                                                                                                                                                    |
| Safety functions                                                                                                                                                                                                                 | Safety systems required by applicable rules for classification. E.g. emergency shutdown, machinery protection, boiler safety, ignition source control, cargo tank overfill protection, gas detection system.                                                                                                                           |
| Lighting                                                                                                                                                                                                                         | System needed to control main, emergency and navigation lights.                                                                                                                                                                                                                                                                        |
| Fire safety                                                                                                                                                                                                                      | Fire detection and alarm system and fixed fire extinguishing system required by applicable statutory regulations or rules for classification.                                                                                                                                                                                          |
| Navigation                                                                                                                                                                                                                       | Navigational systems and equipment required by applicable statutory regulations (e.g. SOLAS, MODU Code, HSLC Code): radar, AIS, GNSS, ECDIS, gyro, heading and bearing indicator, BNWAS, echo sounder, electronic plotting, speed and distance measuring device, tracking aid, rate of turn indicator and transmitting heading device. |
| Internal communication                                                                                                                                                                                                           | Internal communication systems required by applicable statutory regulations. E.g. public announcement, general alarm, two-way voice communication.                                                                                                                                                                                     |
| External communication                                                                                                                                                                                                           | External communication systems required by applicable statutory regulations. E.g. Global maritime distress and safety system.                                                                                                                                                                                                          |
| <sup>1)</sup> Systems providing local, remote or automatic control as well as required monitoring, alarm, and safety functions shall be included in the SuC.<br>Integrated control systems shall be included in the default SuC. |                                                                                                                                                                                                                                                                                                                                        |

# Less systems in scope

Systems that represent negligible cyber risk may be exempted from scope

**E26**  
(cont)

## **6 Risk assessment for exclusion of CBS from the application of requirements**

### **6.1 Requirement**

A risk assessment shall be carried out in case any of the CBSs falling under the scope of applicability of this UR is excluded from the application of relevant requirements. The risk assessment shall provide evidence of the acceptable risk level associated to the excluded CBSs.

## 2.2 Exemption based on negligible risk

### 2.2.1 Exemption of systems from the SuC

A risk assessment shall be carried out if exempting a computer-based system from the security requirements in [4]. The objective is to determine if the system may be considered to represent negligible cyber risk. The risk assessment shall at least consider the criteria listed below:

- The system shall be isolated and not have IP-network communication with other systems (including no remote access solution).
- The system shall be located in area(s) to which physical access is restricted and controlled.
- The system shall not have accessible physical interface ports. Unused interfaces shall be logically disabled. It shall not be possible to mount external devices to the system.
- The system shall not be an integrated control system serving multiple vessel functions, see [Pt.4 Ch.9 Sec.2 \[1.3\]](#).

The following additional criteria should also be considered in the risk assessment:

- The system should not be needed for propulsion and steering, and should not include required safety functions.
- The attack surface for the CBS is minimized, having considered the system's complexity, connectivity, physical and logical access points, including wireless access points.
- Known vulnerabilities, threats, attack surface, and potential impacts deriving from a cyber incident should be considered

The above risk assessment shall be submitted by the integrator as part of document F011, see [Table 6](#)

#### **Guidance note:**

The above items are intended as an approximation to determine the likelihood of an attack for each individual system. See [DNV-RP-0496](#) for further details and guidance. See also IACS UR E26 chapter 6.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

# Asset inventory

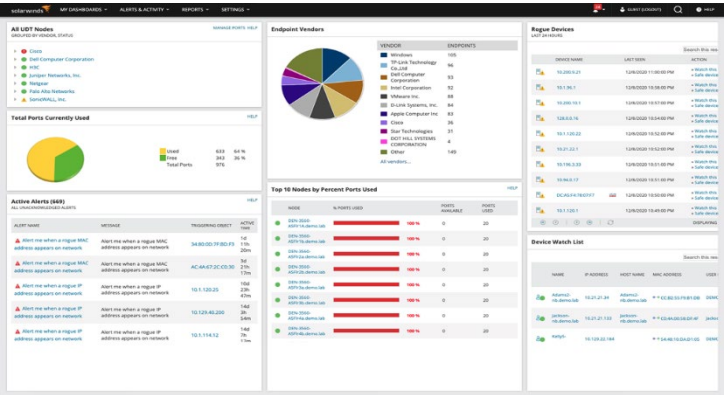
You can't protect your assets unless you know what your assets are

Supplier



IACS E27  
§ 3

Deliver asset inventory as part of system documentation



Source: [www.solarwinds.com](http://www.solarwinds.com)

Shipyards



IACS E26  
§ 4.1.1

Deliver the ship with a complete and updated inventory of hardware and software for the systems in scope.

Owner



IACS E26  
§ 4.1.1

Keep asset inventory updated

DNV



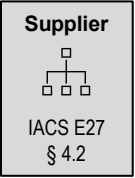
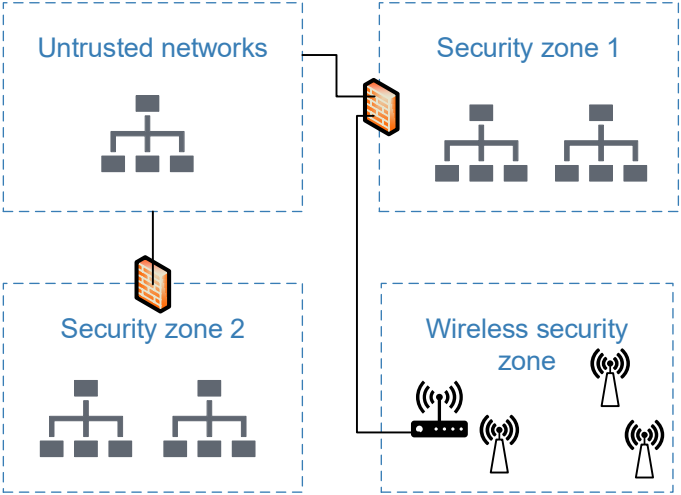
Cyber  
secure  
(Essential)

Aligned

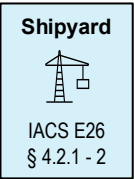
# Security zones

Systems to be grouped in security zones, and zone boundaries (conduits) to be protected.

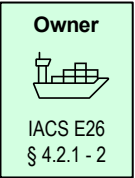
Safety, navigation and wireless systems allocated to separate security zones



Coordinate any communication through conduits with shipyard, note E27 §4.2



Design security zones, control communication through conduits



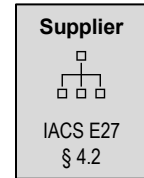
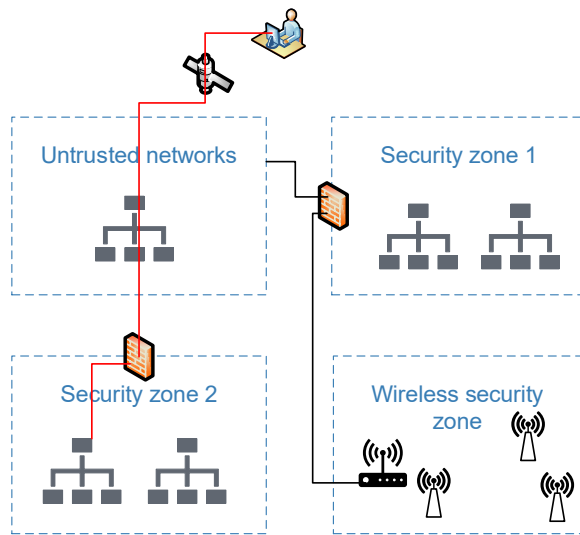
Manage the zones and conduits in operation



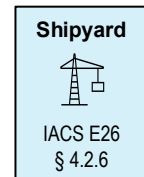
Aligned

# Remote access / untrusted networks

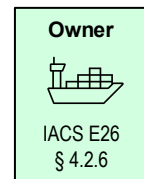
Prevent cyber incidents originating from external/untrusted networks



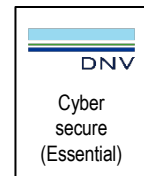
- ✓ Multifactor authentication (SR 1.1 RE1)
- ✓ Endpoint authentication (SR 1.2)
- ✓ Limit consecutive login attempts (SR 1.11)
- ✓ Use notification (SR 1.12)
- ✓ Monitor and control remote access (SR 1.13)
- ✓ Explicit access request approval (SR 1.13 RE1)
- ✓ Remote session termination (SR 2.6)
- ✓ Cryptographic integrity protection (SR 3.1 RE1)
- ✓ Session integrity (SR 3.8 / SR 3.8 RE1)



Shipyard to deliver user manual for remote access, ensure the systems are not exposed to external networks, ensure systems meet E27 § 4.2



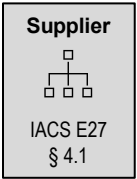
Manage communication between security zones and untrusted networks in operation



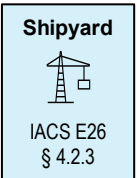
Aligned

# Malicious code

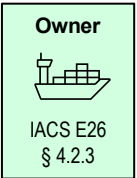
Protect systems against virus, worms, trojan horse, malicious code and other harmful software.



Implement malicious code protection in the system (SR 3.2)



Ensure systems in scope are protected against malicious code



Manage the malware protection in operation

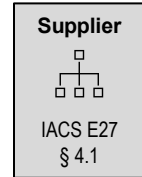


Aligned

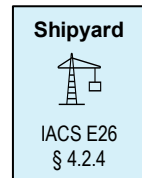


# Access control

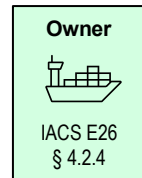
Prevent unauthorized access to the systems



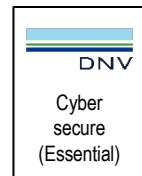
- ✓ Identify and authenticate human users (SR 1.1)
- ✓ Manage user accounts, identifiers and authenticators (SR 1.3, SR 1.4, SR 1.5)
- ✓ Authorize users as per least-privilege principle (SR 2.1)



Ensure logical and physical access control is implemented as per the least privilege principle.



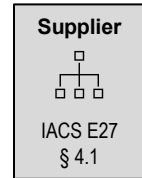
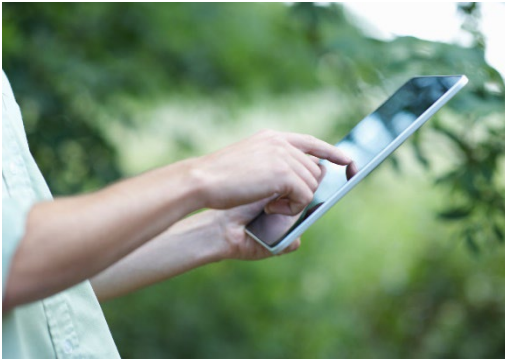
Manage access controls and authorizations in operation



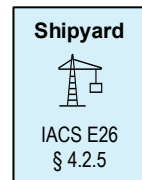
Aligned

# Wireless communication

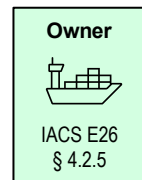
Prevent cyber incidents in wireless networks and propagation to other systems



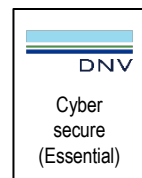
- ✓ Identify and authenticate all users (SR 1.6)
- ✓ Use control according to commonly accepted industry practices (SR 2.2)
- ✓ Protect information integrity and confidentiality (SR 3.1 / SR 4.1)



Ensure wireless networks are separate security zones, and prevent unauthorized access



No specific requirements in operation



Aligned

# Portable and mobile devices

Prevent cyber incidents via portable devices such as laptops, tablets, USB memory sticks, cameras, etc.



**Supplier**

IACS E27  
§ 4.1

Prevent use, restrict transfer of code, or require context-specific authorization (SR 2.3)

**Shipyard**

IACS E26  
§ 4.2.7

Ensure systems meet E27, consider blocking physical interface ports

**Owner**

IACS E26  
§ 4.2.7

Management, awareness, policies and procedures

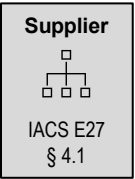
DNV

Cyber  
secure  
(Essential)

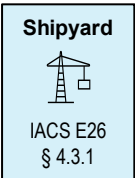
Aligned

# Network monitoring

Detect and prevent abnormal network performance



- ✓ Denial of service protection (SR 7.1)
- ✓ Prevent or control connection of unauthorized devices to the network (SR 2.3)
- ✓ Record security events (SR 2.8) with timestamp (SR 2.11)



Ensure systems meet E27



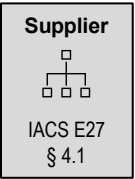
No specific requirements in operation



Aligned

# Diagnostics and testing

Ensure correct operation of security functions



- ✓ Support verification/test of security functions (SR 3.3)
- ✓ Deliver guidelines for configuration (SR 7.6)



Shipyard to ensure systems meet E27.



No specific requirements in operation



Aligned

# Incident response

Support the ship owner's plans for incident response



**Supplier**



IACS E27  
§ 4.1

Ensure systems fail to safe state in the event of cyber incidents (SR 3.6)

**Shipyard**



IACS E26  
§ 4.1 - 4

- ✓ Provide information to owner for incident response plans
- ✓ Ensure required local/backup controls are independent
- ✓ Ensure isolate network segments can be isolated
- ✓ Ensure systems respond to cyber incidents in a safe manner

**Owner**



IACS E26  
§ 4.1 - 4

Develop incident response plan

**DNV**

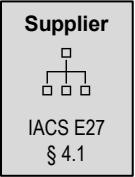
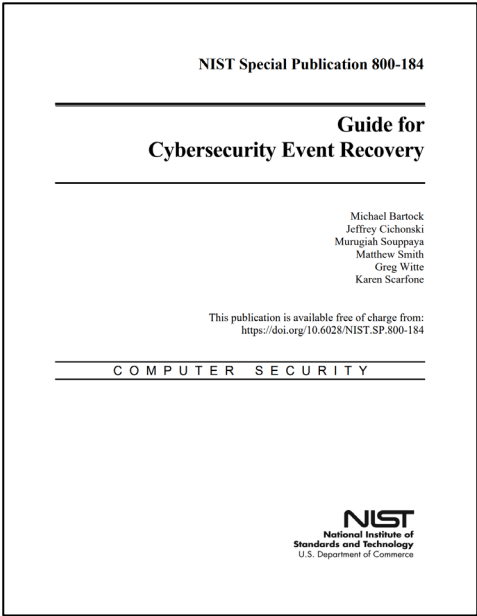
Cyber  
secure  
(Essential)

Aligned

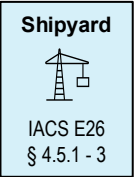


# Business continuity

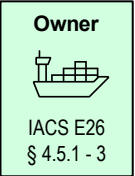
Support the ship owner’s plans for business continuity



Ensure capabilities to backup and recover the systems (SR 7.3 and SR 7.4)



Provide information to owner for business continuity plan (i.e. backup, restore, recovery and reconstitution).

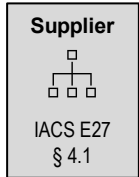


Develop business continuity plan



Aligned

# Additional system hardening



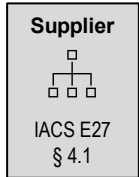
Ensure systems meet additional hardening requirements as per IEC 62443-3-3 SL1

- ✓ Configurable password strength (SR 1.7)
- ✓ Obscure feedback of authenticator information (SR 1.10)
- ✓ Enforce usage restrictions for mobile code (SR 2.4)
- ✓ Session lock (SR 2.5)
- ✓ Storage capacity for audit records (SR 2.9)
- ✓ Alert of audit processing failures (SR 2.10)
- ✓ Protect integrity of transmitted information (SR 3.1)
- ✓ Validate process control inputs (SR 3.5)
- ✓ Protect confidentiality of sensitive information (SR 4.1)
- ✓ Apply commonly accepted industry practice for cryptography (SR 4.3)
- ✓ Allow authorized users read-only access to audit records (SR 6.1)
- ✓ Prevent resource exhaustion by security functions (SR 7.2)
- ✓ Prevent loss of security functions due to single power failure, if applicable (SR 7.5)
- ✓ Prevent use of unnecessary functions (SR 7.7)



Aligned

# Secure product development lifecycle



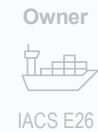
Demonstrate compliance with a selected subset of the secure product lifecycle requirements in IEC 62443-4-1

- ✓ Protect private keys used for code signing (SM-8)
- ✓ Inform end users about security updates for the system (SUM-2)
- ✓ Demonstrate that the system is compatible with updated versions of acquired products (SUM-3)
- ✓ Make security updates available to end users, and ensure the end user can verify the authenticity of the updated software (SUM-4)
- ✓ System-specific defence in depth strategy (SG-1)
- ✓ Site-specific defence in depth guidelines (SG-2)
- ✓ Security hardening guidelines (SG-3)



Rules will be adjusted. With current rules, an additional audit of product suppliers will be added to the type approval scheme.

## 63



- Document and submit security management processes
- Implement security management on board
- Demonstrate management processes in annual surveys
- Demonstrate/test security controls in complete survey

# Cyber security type approval

# Certification of Materials and Components (CMC) related cyber security class activities

Supplier

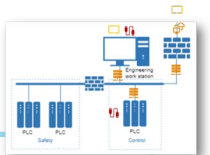
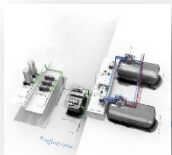


IACS E27

## Mandatory product certification



CS Type approval?



### Full Case-by-case w/o TA

All systems in scope and not deemed negligible risk (no network, SW, updates, access, ...) will require

- i. **CS system design approval** – DNV approval centre
- ii. **CS system survey** – by local DNV office

### Reduced Case-by-case with TA

All systems in scope and not deemed negligible risk (no network, SW, updates, access, ...) will require

- i. **Reduced CS system design approval** – confirm applicability of TA/configuration/supplier test report
- ii. **CS system survey** – omitted





# Activities, suppliers

## Preparations (now)

Determine if system is in scope

Implement required security capabilities

Establish Secure Development Lifecycle processes

Evaluate feasibility of Type approval

## For each delivery - if system is type approved

Submit reduced set of documents to class society

- F030 - System topology
- F071 - Inventory list
- F262 - Test report or declaration of hardening and security configuration

Obtain Product certificate

## For each delivery - if system is not type approved

Submit complete set of documents to class society

- F030 - System topology
- F071 - Inventory list
- F021 - Description of security capabilities
- F252 - Procedure for test of security capabilities
- F259 - Description of secure development lifecycle processes
- F141 - Guidelines for hardening and configuration of security capabilities
- F142 - Information supporting incident response and recovery
- F253 - Procedure for verification of security functions on board
- I320 - Software change handling procedure.

Factory acceptance test

Obtain Product certificate

# System vs. Component



The technical security capabilities in IEC 62443-3-3, DNV rules, and IACS UR E27 are intended for systems.



System: More than one component in IEC 62443-4-2 forming a system designed to serve an intended OT-function onboard. (Host device, embedded device, network device, software application)



For most SP1-requirements, it is not considered possible to verify compliance of a system unless all components in the system have the required security capability, as applicable.

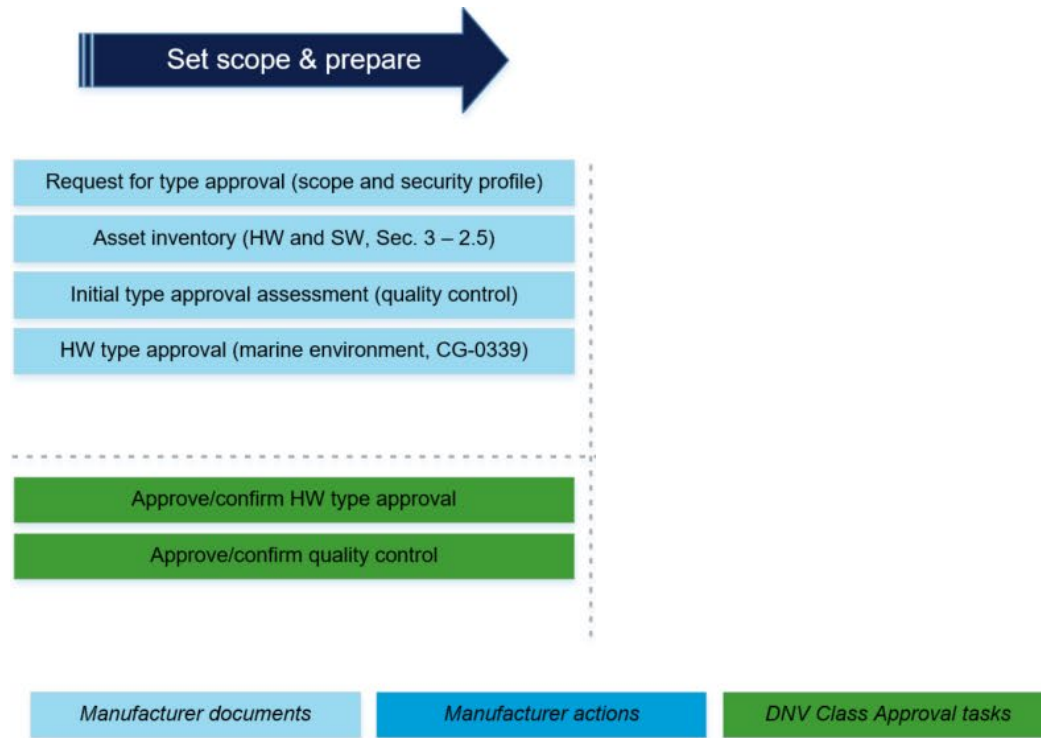


In the “Description of security capabilities” and “Test procedure”, address each unique type of component in the system.



Nearly all required security capabilities are software functions (exceptions? SR 5.1 RE1, SR 5.2 RE2, SR 7.5)

# Type approval confirms design, implementation, testing & life cycle of cyber security barriers



## 5. Security capabilities (Suppliers)

### Identification and authentication

SR 1.1 Identification and authentication  
SR 1.3 Account management  
SR 1.4 Identifier management  
SR 1.5 Authenticator management  
SR 1.6 Wireless access management  
SR 1.7 Strength of passwords  
SR 1.10 Authenticator feedback

### Use control

SR 2.1 Authorization enforcement  
SR 2.2 Wireless use control  
SR 2.3 Portable and mobile devices  
SR 2.4 Mobile code  
SR 2.5 Session lock  
SR 2.8 Auditable events  
SR 2.9 Audit storage capacity  
SR 2.10 Audit processing failures  
SR 2.11 Timestamps (SL2)

### System integrity

SR 3.1 Communication integrity  
SR 3.2 Malicious code protection  
SR 3.3 Security functionality verification  
SR 3.6 Deterministic output

### Data confidentiality

SR 4.1 Confidentiality of “sensitive” information  
SR 4.3 Use of cryptography

### Restricted data flow (UR E26)

SR 5.1 Network segmentation  
SR 5.1 RE1 Physical network segmentation  
SR 5.2 Zone boundary protection  
SR 5.2 RE1 Deny by default  
SR 5.3 General purpose P2P  
SR 5.4 Application partitioning

### Timely response to events

SR 6.1 Audit log accessibility

### Resource availability

SR 7.1 Denial of service protection  
SR 7.2 Resource management  
SR 7.3 Backup  
SR 7.4 Restore  
SR 7.5 Emergency power  
SR 7.6 Configuration guidelines  
SR 7.7 Least functionality

### Interfaces with other/untrusted networks

SR 1.1 RE2 Multifactor authentication (SL3)  
SR 1.2 Endpoint authentication ((SL2)  
SR 1.11 Unsuccessful login attempts (SL1)  
SR 1.12 System use notification (SL1)  
SR 1.13 Access via untrusted networks (SL1)  
SR 1.13 RE1 Explicit access approval (SL2)  
SR 2.6 Remote session termination (SL2)  
SR 3.1 RE1 Cryptographic integrity prot. (SL3)  
SR 3.5 Input validation (SL1)  
SR 3.8 Session integrity (SL2)  
SR 3.8 RE1 Invalidation of session ID (SL3)

### 5.10 SM-8: Controls for private keys

#### 5.10.2 Rationale and supplemental guidance

Private keys are the root of trust, so they require extra protection to ensure that they are not stolen or modified.

Required by [DNV-RU-SHIP Pt.6 Ch.5 Sec.21](#)  
– specified in IEC 62443-4-1

See e.g., OWASP [https://cheatsheetseries.owasp.org/cheatsheets/Key\\_Management\\_Cheat\\_Sheet.html](https://cheatsheetseries.owasp.org/cheatsheets/Key_Management_Cheat_Sheet.html)



### OWASP Cheat Sheet Series

This Key Management Cheat Sheet provides developers with guidance for implementation of cryptographic key management within an application in a secure manner. It is important to document and harmonize rules and practices for:

1. Key life cycle management (generation, distribution, destruction)
2. Key compromise, recovery and zeroization
3. Key storage
4. Key agreement

## 6. Secure development lifecycle processes (Suppliers)

### 11.3 SUM-2: Security update documentation

#### 11.3.2 Rationale and supplemental guidance

This process is required to ensure that security patches are documented to allow approved patches to be installed and non-approved patches to be remediated.

Having this process means that the product supplier provides or otherwise makes documentation available that identifies and describes applicable security patches, how to install approved patches, how to determine patch status (whether a patch has been applied) of components and how to mediate non-approved patches. See the patch management requirements of IEC 62443-2-4 for more information.

### 11.4 SUM-3: Dependent component or operating system security update documentation

#### 11.4.2 Rationale and supplemental guidance

End users are hesitant to install software in an IACS that might upset operations. As a result, vendors need to provide information to the users about whether a particular security update of the operating system is compatible with the product.

### 11.5 SUM-4: Security update delivery

#### 11.5.2 Rationale and supplemental guidance

This process is required to ensure that product users can obtain applicable security patches for the product in a timely manner and to reduce the possibility that the security patches are fraudulent (see Clause 11).

Having this process means that the product supplier provides a mechanism or technique that allows product users to verify the authenticity of patches. Concurrent release of patches for all supported versions can reduce the time window between awareness of the vulnerability and the availability of patches.

Required by [DNV-RU-SHIP Pt.6 Ch.5 Sec.21](#)  
– specified in IEC 62443-4-1

### 12.2 SG-1: Product defense in depth

#### 12.2.2 Rationale and supplemental guidance

This process is required to ensure that documentation for the defense in depth strategy is produced to support hardening of the product at the customer site. Such documentation is required by IEC 62443-2-4, that defines security requirements for IACS installation and maintenance service providers.

Having this process means that the product supplier documents various aspects of the defense in depth strategy necessary to harden the product during installation and keep it hardened during its lifetime of use. Aspects of the defense in depth strategy to be documented include the residual threats that are expected to be present and capable of attacking the product, the security capabilities of the product to safeguard it against these threats and any compensating security controls/mitigations that can be used with the product to further protect the product.

### 12.3 SG-2: Defense in depth measures expected in the environment

#### 12.3.2 Rationale and supplemental guidance

This process is required to ensure that documentation for the defense in depth strategy is produced to support hardening of the product at the customer site. Such documentation is required by IEC 62443-2-4, that defines security requirements for IACS installation and maintenance service providers.

Having this process means that the product supplier documents various aspects of the defense in depth strategy necessary to harden the product during installation and keep it hardened during its lifetime of use.

Required by [DNV-RU-SHIP Pt.6 Ch.5 Sec.21](#)  
– specified in IEC 62443-4-1



### 12.4 SG-3: Security hardening guidelines

#### 12.4.2 Rationale and supplemental guidance

This process is required to ensure that instructions that describe how to harden the product and keep it hardened are documented. Such documentation is required by IEC 62443-2-4 that defines security requirements for IACS installation and maintenance service providers.

Having this process means that the product supplier creates user documentation that provides directions for hardening the product during installation and for keeping it hardened during the lifetime of the product use. This requirement recognizes that the security policies and requirements for customer sites are often different, and as a result, instructions for securely integrating the product into the customer site, configuring it appropriately and maintaining its security are necessary.

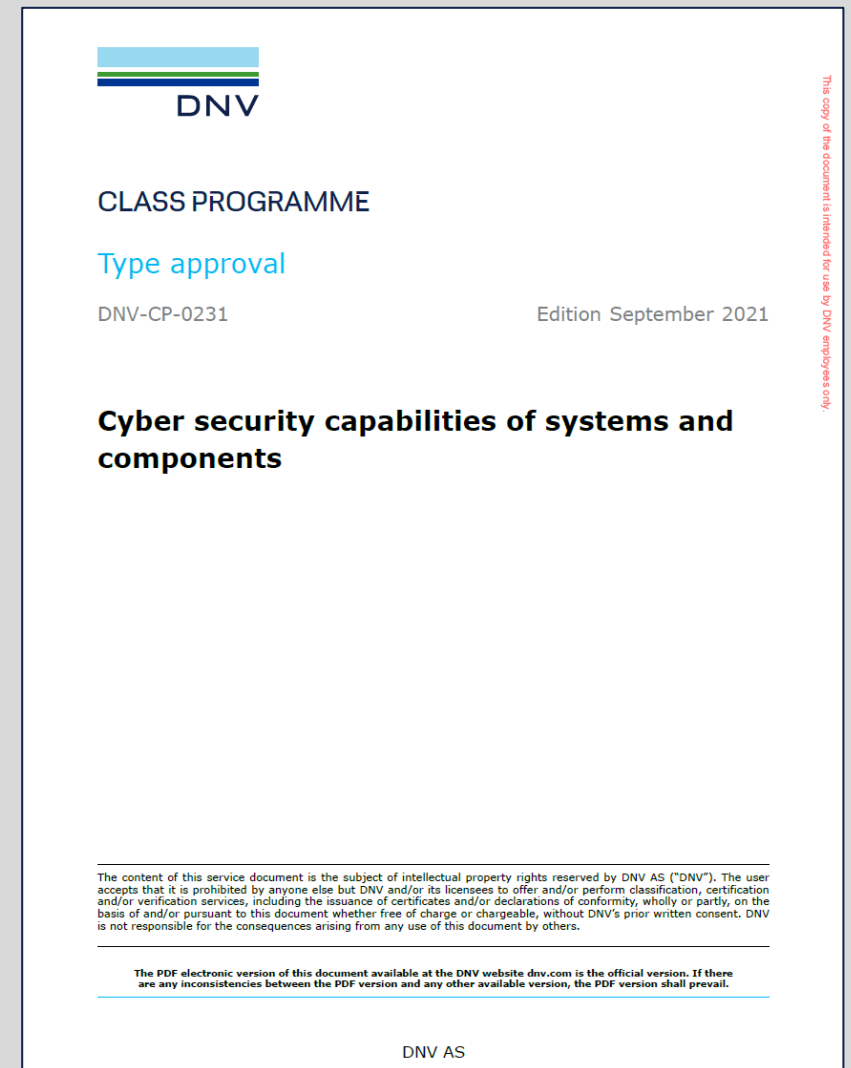
Required by [DNV-RU-SHIP Pt.6 Ch.5 Sec.21](#)  
– specified in IEC 62443-4-1

See also SR 7.7 – Least functionality

See DNV class programme [DNV-CP-0231](#)

### TA process

1. RFQ - Supplier submits request for quotation to DNV local station
2. Quotation/contract - DNV local station provides quotation to supplier
3. Initial TA assessment – QA assessment by DNV local station, if required
4. Documentation - Supplier submits documentation for TA
5. Type test - Supplier invites for type testing
6. TA certificate - TA certificate is issued.



### Supplier submits request for quotation to DNV local station

RFQ should include:

- Scope: TA of product/system in accordance with DNV rules for class notation Cyber secure(Essential)
- Brief functional description of the system
- List of devices in system
- Topology diagram of the system
- Production sites from where the type approved product will be delivered

#### 1. RFQ

2. Contract
3. Initial TA
4. Documentation
5. Type test
6. TA certificate



### APPLICATION FOR TYPE APPROVAL OF INSTRUMENTATION AND AUTOMATION EQUIPMENT

This form should be filled in by the applicants applying for DNV Type Approval in accordance with the DNV rules. Send the completed form to your local DNV office.

**Particulars of Application**

Manufacturer: \_\_\_\_\_

Manufacturer address: \_\_\_\_\_

Place of manufacture: \_\_\_\_\_

Company applying for Type Approval: \_\_\_\_\_

Applicant is authorised by the manufacturer to act on its behalf. Authorisation: ☐ Enclosed

Type Approval of: \_\_\_\_\_

Standards: ☐ RU-SHIP (DNV Rules for Classification of Ships)  
☐ RU-HSLC (DNV Rules for Classification of High Speed Light Crafts)  
☐ RU-OU (DNV Rules for Classification of Offshore Units)  
☐ Other: \_\_\_\_\_

**Type Approval shall cover**

☐ Hardware ☐ Software

**Environmental tests:**

☐ Have been carried out ☐ Have not been carried out

Name of laboratory: \_\_\_\_\_

Address of laboratory: \_\_\_\_\_

**EMC tests:**

☐ Have been carried out ☐ Have not been carried out

Name of laboratory: \_\_\_\_\_

Address of laboratory: \_\_\_\_\_

**The equipment for testing is:**

☐ Production sample ☐ Prototype model



Form code: 86.02a Revision: 2022-12 www.dnv.com Page 1 of 4

### DNV local station provides quotation and contract to supplier

- DNV approval center will make quotation based on RFQ
- DNV local station will provide quotation and contract to supplier
- The contract is called SFA (short form agreement) and will include the quotation as attachment
- The contract includes standard NDA for certification of materials and components
- Contract to be signed by both parties

1. RFQ
2. **Contract**
3. Initial TA
4. Documentation
5. Type test
6. TA certificate

**DNV**

**CERTIFICATION OF MATERIALS AND COMPONENTS**

Short Form Agreement ("Agreement")

DNV Doc No: 32928870/DNV

**Section I - Cover letter**

**"Customer"**

Legal entity: [REDACTED]  
Legal entity VAT no: [REDACTED]  
Contact person: [REDACTED]  
Business address: [REDACTED]  
Customer no.(id): [REDACTED]  
Phone/mail: / /  
Invoicing address: [REDACTED]  
Is purchase order no. required to process invoice: ☐ No ☐ Yes; P.O. no: [REDACTED]

**"DNV"**

Legal entity: **DNV AS**  
Legal entity VAT no: [REDACTED]  
Contact person: **Omberg, Knut**  
Business address: **Veritasveien 1, 1363 Høvik, Norway**  
DNV order no: **262.1-039424-1-J-10**  
Phone/mail: **+4745194764 / Knut.Omberg@dnv.com**

**Work/project**

Project name: [REDACTED]  
Commencement date: 1 Contract end date: 1  
Work location: [REDACTED] Project number (if applicable): **262.1-039424-1**  
Applicable rules: **Choose a rule set**

Form code: AGR 301 Revision: 2023-01 www.dnv.com Page 1 of 10

## 7. Type approval (initial TA assessment)

### Quality system

- ✓ Supplier to arrange initial TA assessment with DNV local station - for each production site
- ✓ A general QA audit, same for type approval of all software-based systems
- ✓ Objective: to ensure consistent production of the type approved product

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6. TA certificate

| Initial Assessment                                                                                                                                                                                                  |                          |                          |                          |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Assessment Item                                                                                                                                                                                                     | Yes                      | No                       | N/A                      | Remarks                  |
| <b>General Items</b>                                                                                                                                                                                                |                          |                          |                          |                          |
| Is product certification performed regularly and at least every 3 month?                                                                                                                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Is an MSA established and retention assessments carried out at least annually?                                                                                                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Has the company a quality system certified to ISO 9001 or equivalent by an accredited body?                                                                                                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| <b>Checklist items</b>                                                                                                                                                                                              |                          |                          |                          |                          |
| Are procedures for inspection and control of deliveries from subcontractors and outsourced production established and followed?                                                                                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Are procedures for quality control of the products during production established and followed?                                                                                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Are procedures for final inspection and testing of the product established and followed?                                                                                                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Are inspection and test results recorded and documented?                                                                                                                                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Is production equipment functioning satisfactorily and calibrated?                                                                                                                                                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Is production, testing and quality control performed by competent personnel?                                                                                                                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Is the production performed on the basis of design documentation that shall be the basis for type approval?                                                                                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Has the company the property right in the design documentation?                                                                                                                                                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Does the company take the responsibility for the complete product and its compliance to the requirements?                                                                                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Is the product manufactured by one or more companies? If yes, the list of companies shall be reported                                                                                                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Has the company procedures that ensure required quality and consistent production from all manufacturers?                                                                                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Is sufficient identification and traceability ensured through the marking of the product?                                                                                                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Has the company been informed about the difference between TA and product certification and the requirements for its product?                                                                                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Has DNV received an updated asbestos free declaration (applicable for changes to the product, when adding new products to the TAC, when sub manufacturers or sub suppliers have changed and when renewing the TAC)? | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| <b>Conclusion</b>                                                                                                                                                                                                   |                          |                          |                          |                          |

May be exempted for production sites that hold valid DNV TA certificates

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### Supplier submits documentation for TA via DNV Maritime Customer Portal

- Documentation is described in [DNV-RU-SHIP Pt.6 Ch.5 Sec.21](#) and [DNV-CP-0231](#)
- DNV-CP-0231 will be updated to improve description of
  - Configuration information (SR 7.6)
  - SDL requirements (SM-8, SUM-2, SUM-3, SUM-4, SG-1, SG-2, SG-3)
  - Supporting information (e.g., documents supporting owner's incident response and recovery)
- Contract includes two transmittals of documents

| The following information shall be submitted for approval:    |                                                                                           | Ref. <a href="#">Pt.6 Ch.5</a> | Ref. <a href="#">DNV-CP-0231</a> |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------|----------------------------------|
| F071                                                          | Asset inventory                                                                           | Sec.21 Table 8                 | Sec.3 [5.5]                      |
| I030                                                          | System topology                                                                           | Sec.21 Table 8                 |                                  |
| F021                                                          | Description of security capabilities                                                      | Sec.21 Table 10                | Sec.3 [5.2]                      |
| Z252                                                          | Procedure for type testing                                                                | Sec.21 Table 10                | Sec.3 [5.3]                      |
| Z259                                                          | Description of SDL processes                                                              | Sec.21 Table 10                |                                  |
| The following information shall be submitted for information: |                                                                                           | Ref. <a href="#">Pt.6 Ch.5</a> | Ref. <a href="#">DNV-CP-0231</a> |
| F141                                                          | Guidelines for security configuration                                                     | Sec.21 Table 10                | Sec.3 [5.4]                      |
| Z251                                                          | Procedure for security functionality verification                                         | Sec.21 Table 10                |                                  |
| Z262                                                          | Templates for documentation of project specific deliveries (incl. security configuration) | Sec.21 Table 9                 |                                  |
|                                                               | Modification strategy                                                                     |                                | Sec.3 [5.6]                      |
|                                                               | Change management procedure                                                               |                                | Sec.3 [5.7]                      |
| Additional supporting information may be requested            |                                                                                           |                                |                                  |

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### Supplier invites for type testing

- Type testing to be witnessed by DNV approval engineer
- Type testing to be done after documents have been approved, including test procedure
- Carried out by supplier at supplier premises
- Remote survey may be possible
- Test witnessing is based on hourly rates, not covered by contract price





### TA certificate is issued

- Certificate is issued after successful document assessment and type testing
- Existing TA certificate for the same product with 2 year validity may be extended
- Published on <https://approvalfinder.dnv.com/>
- TA certificate:
  - Validity period 2 years
  - *“Found to comply with DNV rules for classification – Ships Pt.6 Ch.5 Sec.21 Cyber security”*
  - *“Covers security capabilities in accordance with DNV security profile 1 and IACS UR E27”*
  - *Approval conditions:* For each delivery, documentation to be submitted demonstrating that the delivered product is covered by the TA certificate
  - *Application/Limitation:* Any special circumstances / compensating countermeasures to be verified in approval or survey for each delivery

### CLASS PROGRAMME

#### Type approval

DNV-CP-0231

## Cyber security capabilities of systems and components

# Thank you very much for you attention!

Jarle Coll Blomhoff, Head of Section Digital Ship Systems  
DNV Ship Classification

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